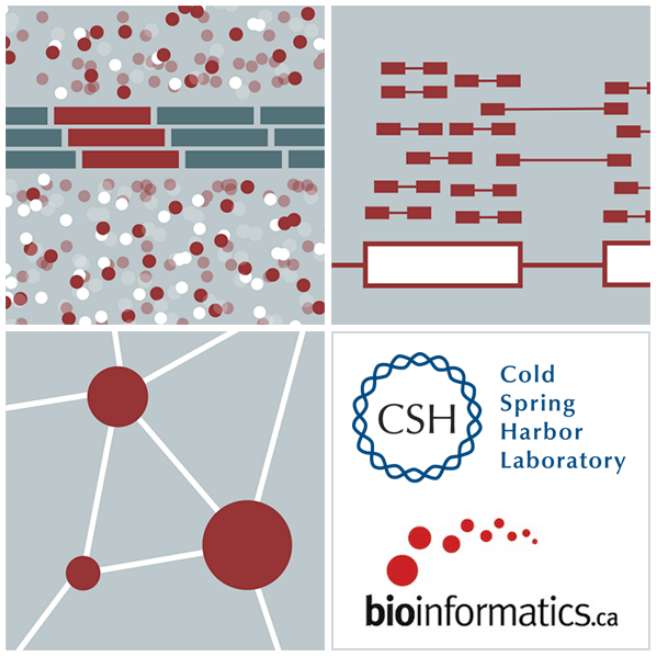
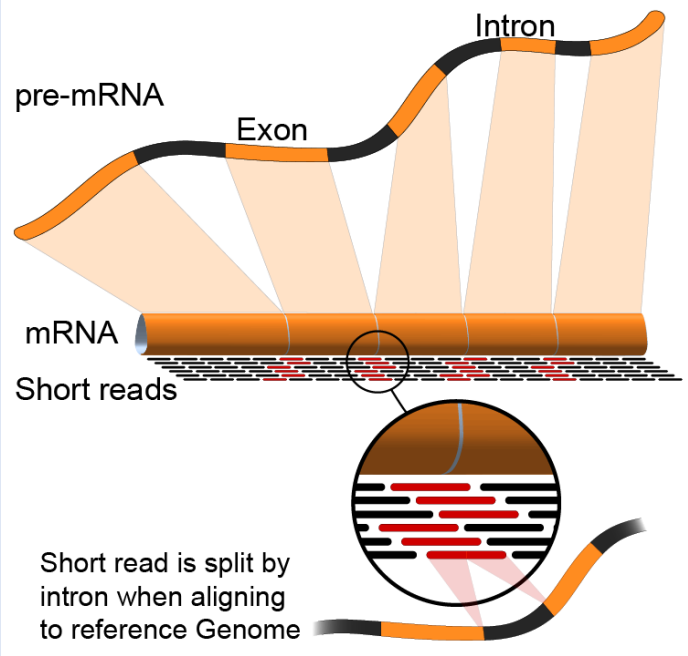
Cold Spring Harbor Laboratory

Introduction to IGV
The Integrative Genomics Viewer

Obi Griffith, Malachi Griffith,
Charlz Jerold, Allen Li, Jennie Yao, Isabel Risch

Advanced Sequencing Technologies & Bioinformatics Analysis November 8-20, 2026





Washington University in St. Louis
School of Medicine

Visualization Tools in Genomics

- there are **over 40 different genome browsers**, which to use?

- depends on
  - task at hand
  - kind and size of data
  - data privacy

HT-seq Genome Browsers



Integrative Genome Viewer



UCSC Genome Browser
Cancer Genome Browser



Trackster (part of Galaxy)

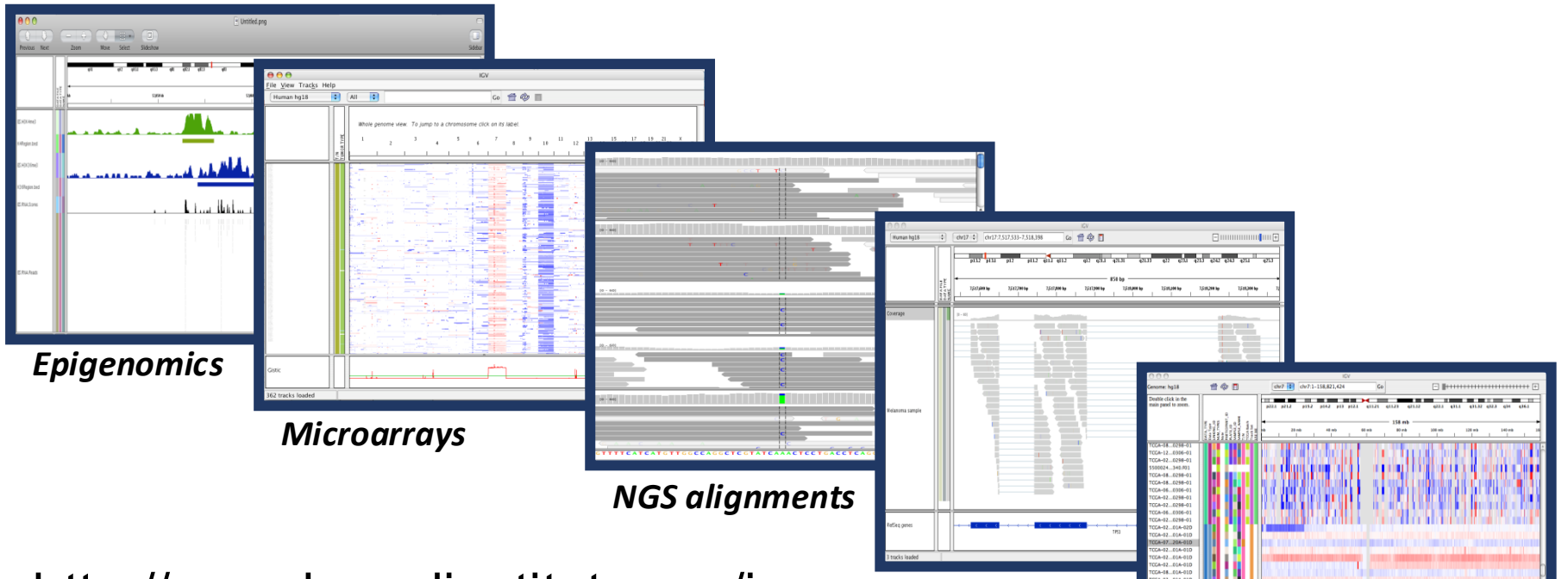


Savant Genome Browser

- task at hand : visualizing HT-seq reads, especially good for inspecting variants
- kind and size of data : large BAM files, stored locally or remotely
- data privacy : run on the desktop, can keep all data private
- UCSC Genome Browser has been retro-fitted to display BAM files
- Trackster is a genome browser that can perform visual analytics on small windows of the genome, deploy full analysis with Galaxy

Integrative Genomics Viewer (IGV)

Desktop application for the interactive visual exploration of integrated genomic datasets



http://www.broadinstitute.org/igv

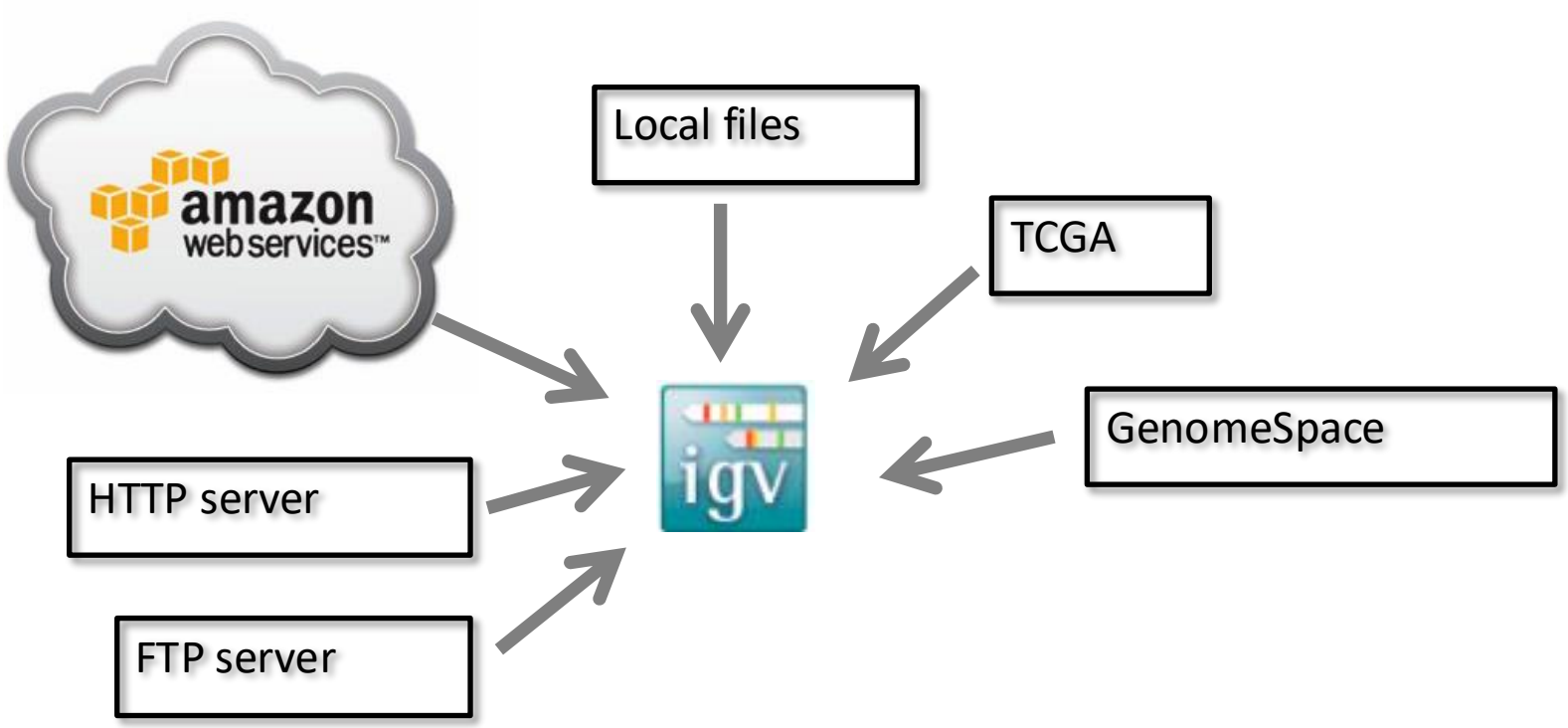
>85,000 registrations (2014)

Features

With IGV you can…

- Explore large genomic datasets with an intuitive, easy-to-use interface.
- Integrate multiple data types with clinical and other sample information.
- View data from multiple sources:
  - local, remote, and “cloud-based”.
- Automation of specific tasks using command-line interface

IGV data sources



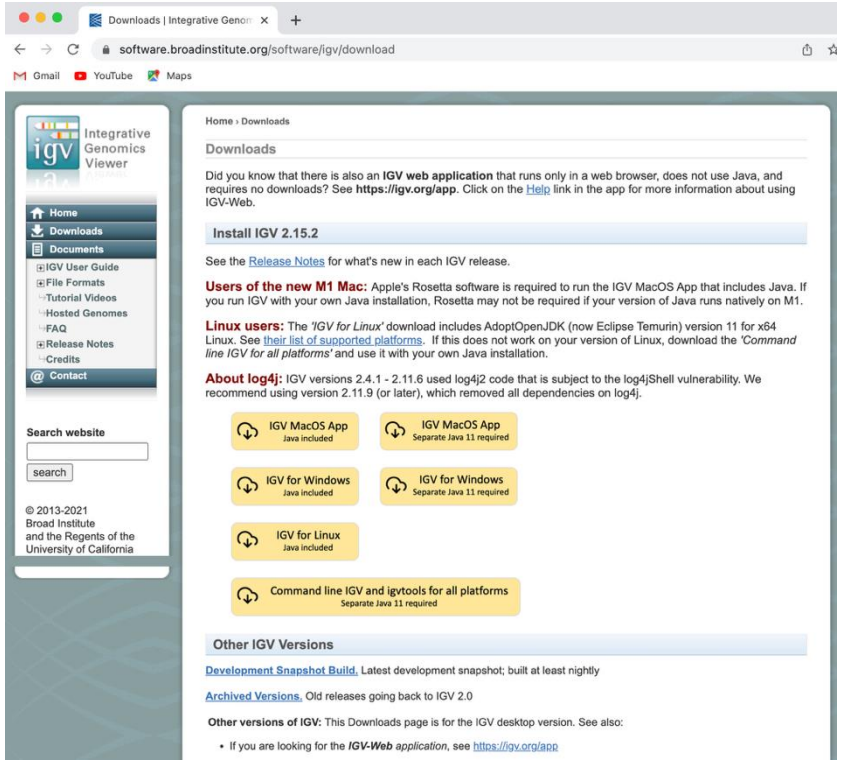
- View **local** files without uploading.
- View **remote** files without downloading the whole dataset.

Using IGV: the basics

- Launch IGV
- Select a reference genome
- Load data
- Navigate through the data
  - WGS data
    - SNVs
    - structural variations

Launch IGV

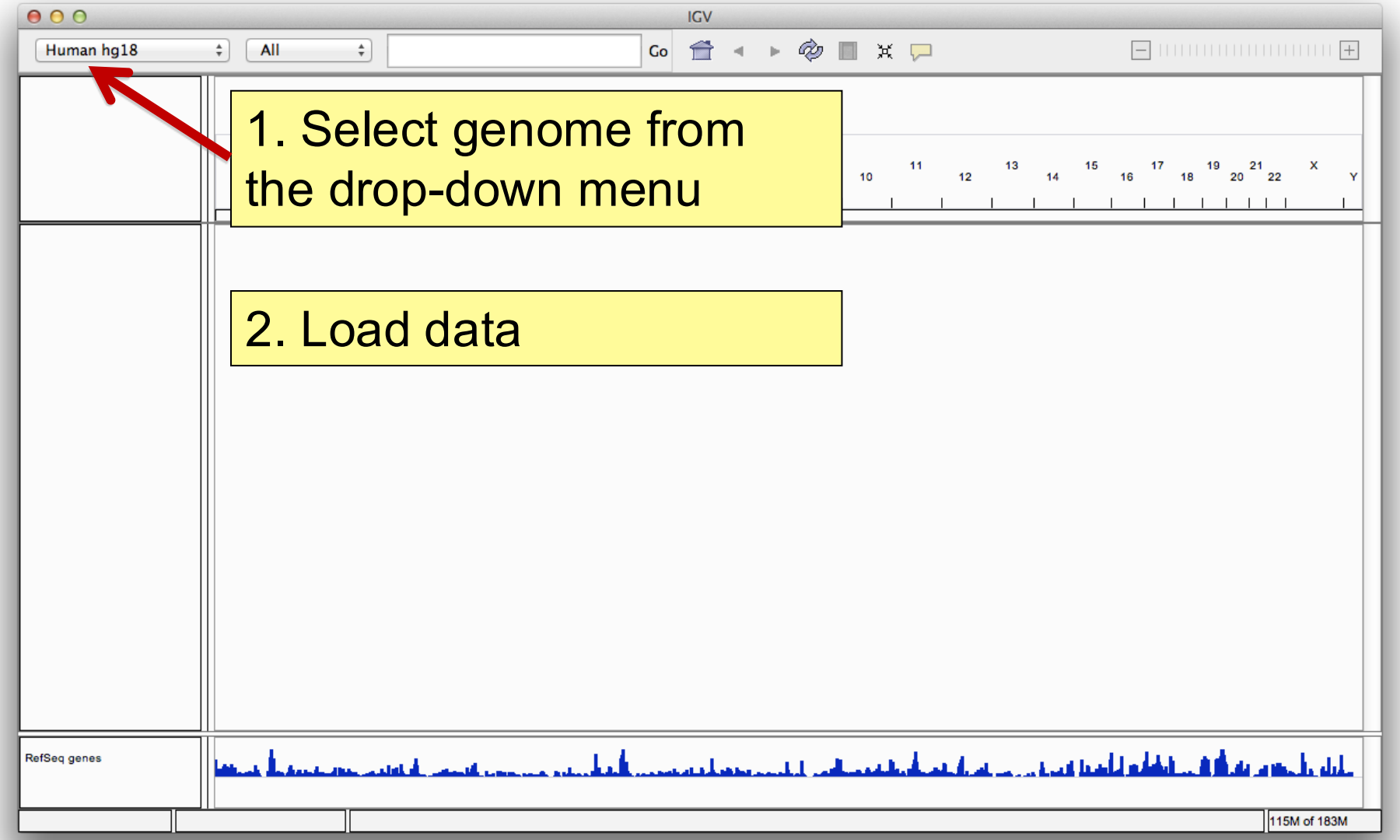

Integrative Genomics Viewer
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Integrative Genomics Viewer
IGV
IGV desktop application
IGV Web App
IGV in a web browser
Juicebox Web
Hi-C contact map viewer

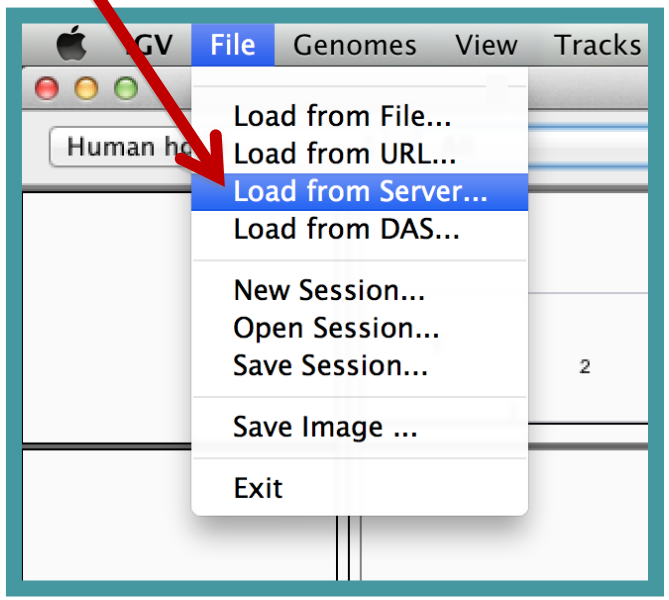

igv

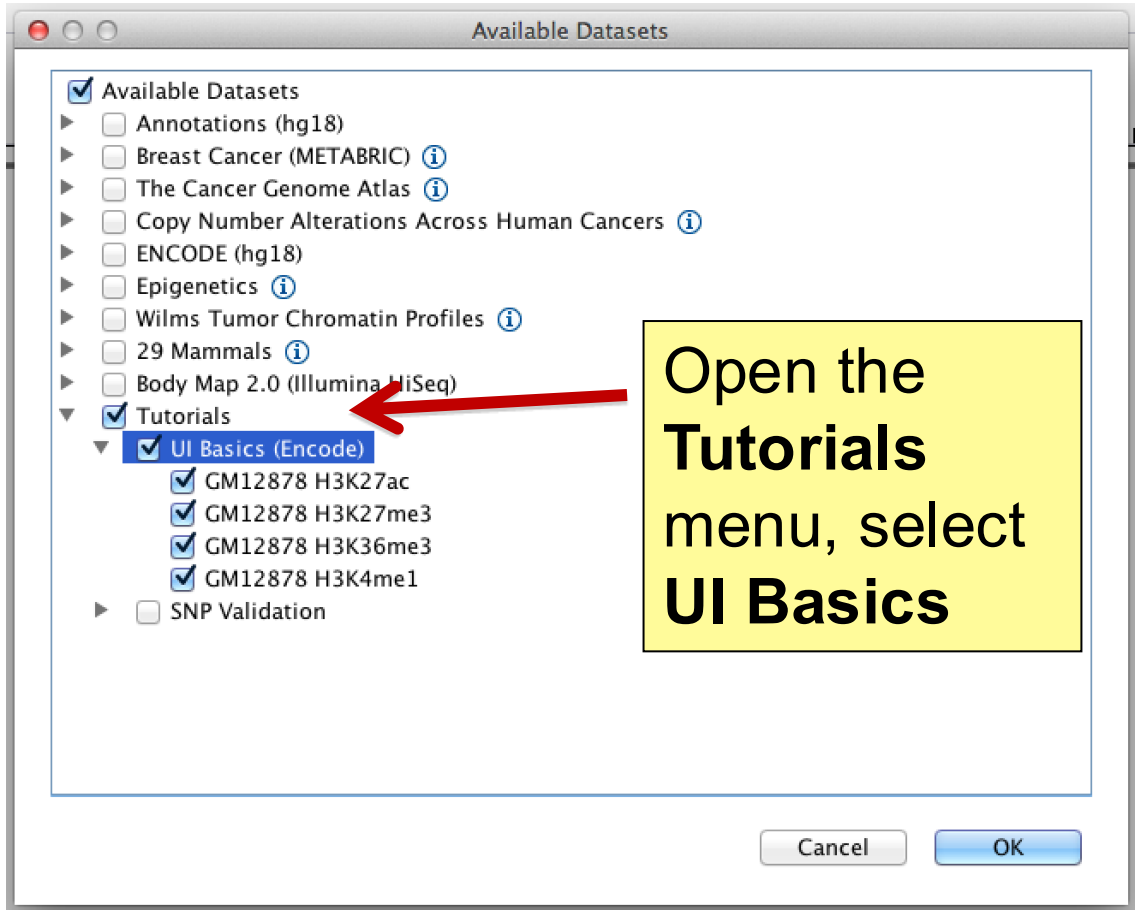
Select reference genome



Load data

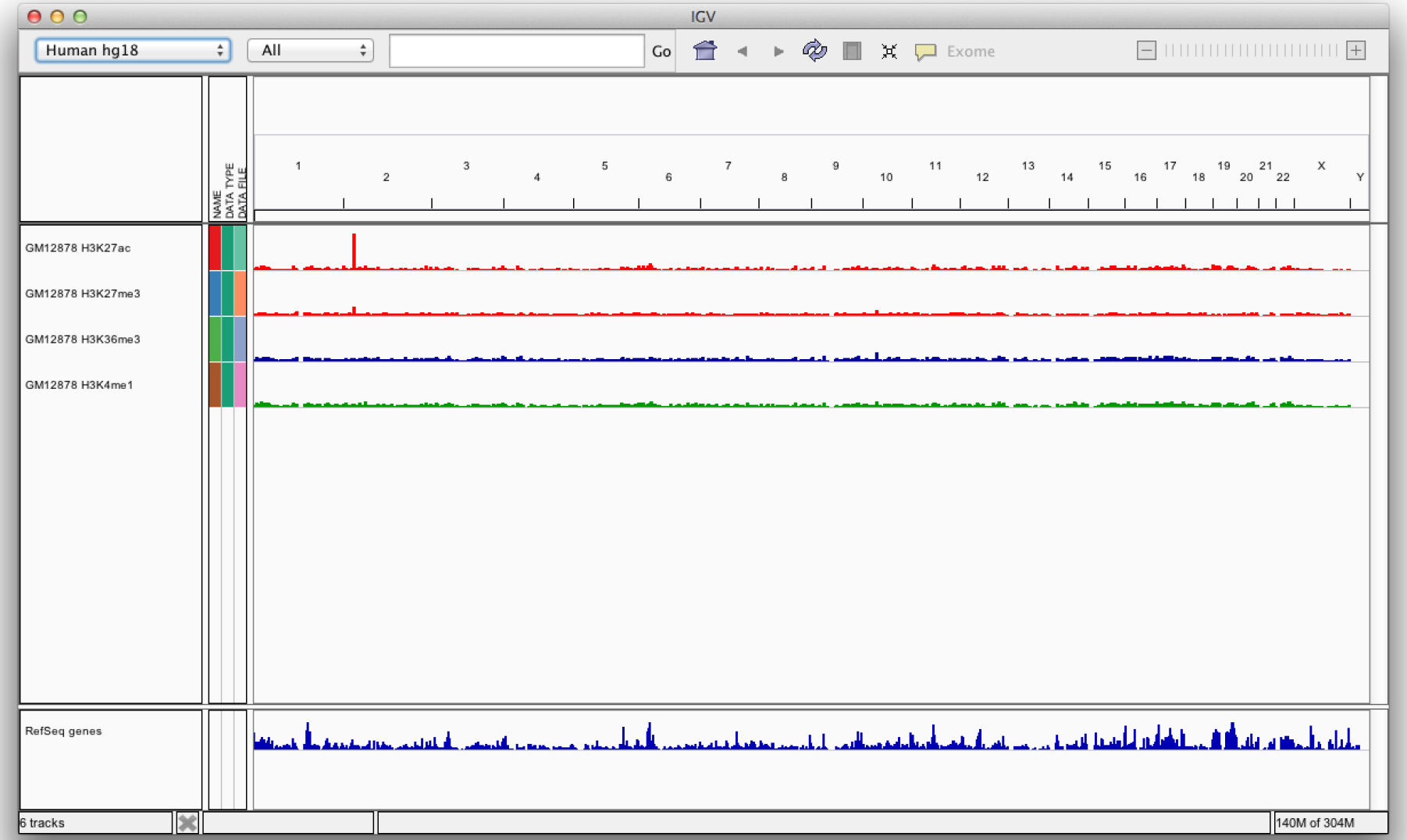
Select **File > Load from Server…**



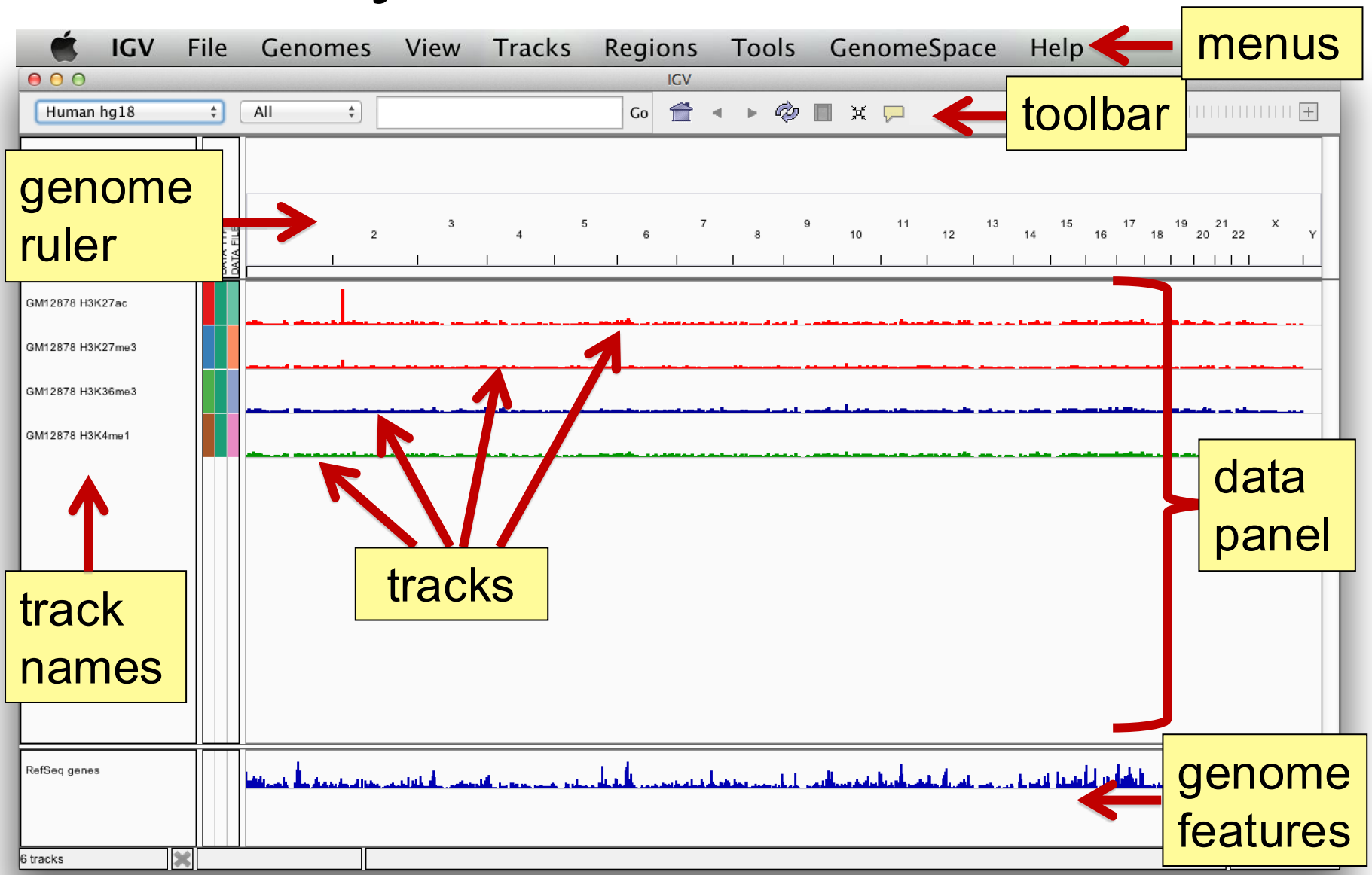


Open the **Tutorials** menu, select **UI Basics**

Screen layout



Screen layout



File formats and track types

- The **file format** defines the track type.
- The **track type** determines the display options

- For
  - BAM
  - BED
  - BEDPE
  - BedGraph
  - bigBed
  - bigWig
  - Birdsuite Files
  - broadPeak
  - CBS
  - Chemical Reactivity Probing Profiles
  - chrom.sizes
  - CN
  - Custom File Formats
  - Cytoband
  - FASTA
  - GCT
  - CRAM
  - genePred
  - GFF/GTF
  - GISTIC
  - Goby
  - GWAS
  - IGV
  - LOH
  - MAF (Multiple Alignment Format)
  - MAF (Mutation Annotation Format)
  - Merged BAM File
  - narrowPeak
  - PSL
  - RES
  - RNA Secondary Structure Formats
  - SAM
  - Sample Info (Attributes) file
  - SEG
  - TDF
  - Track Line
  - Type Line
  - VCF
  - WIG

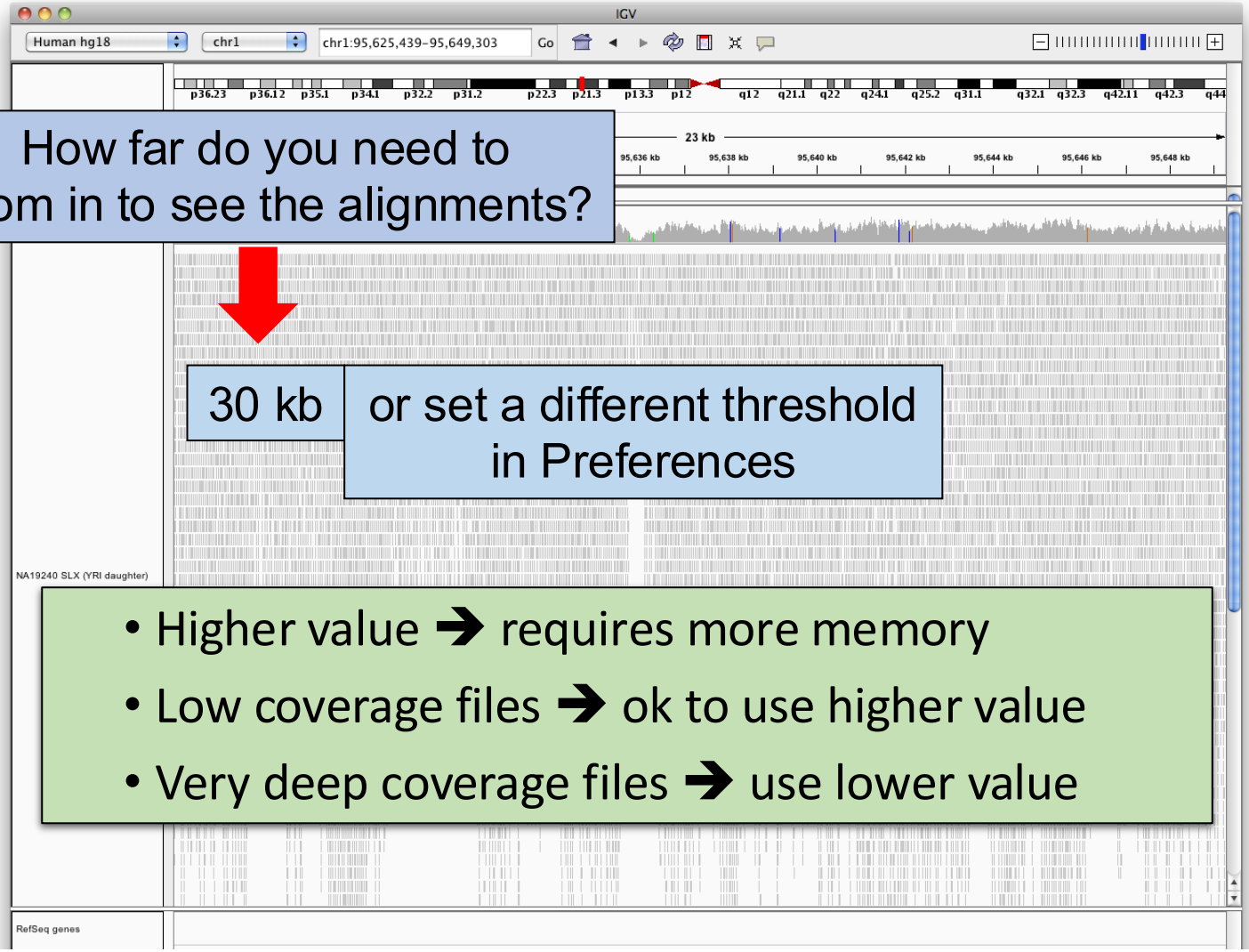
ftware.broadinstitute.org/software/igv/FileFormats

Viewing alignments

Whole chromosome view

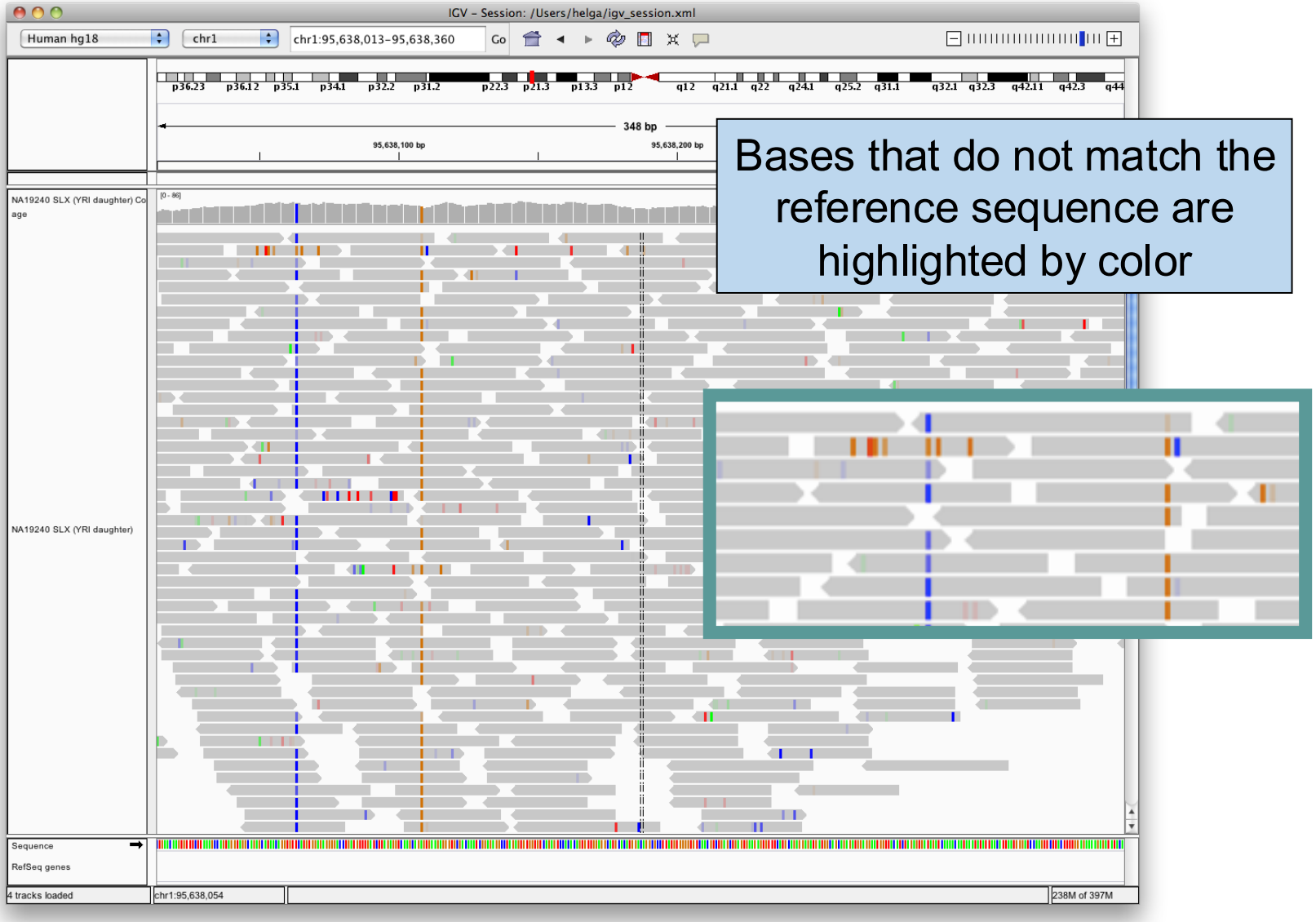


Viewing alignments – Zoom in



- Higher value ➔ requires more memory
- Low coverage files ➔ ok to use higher value
- Very deep coverage files ➔ use lower value

Viewing alignments – Zoom in



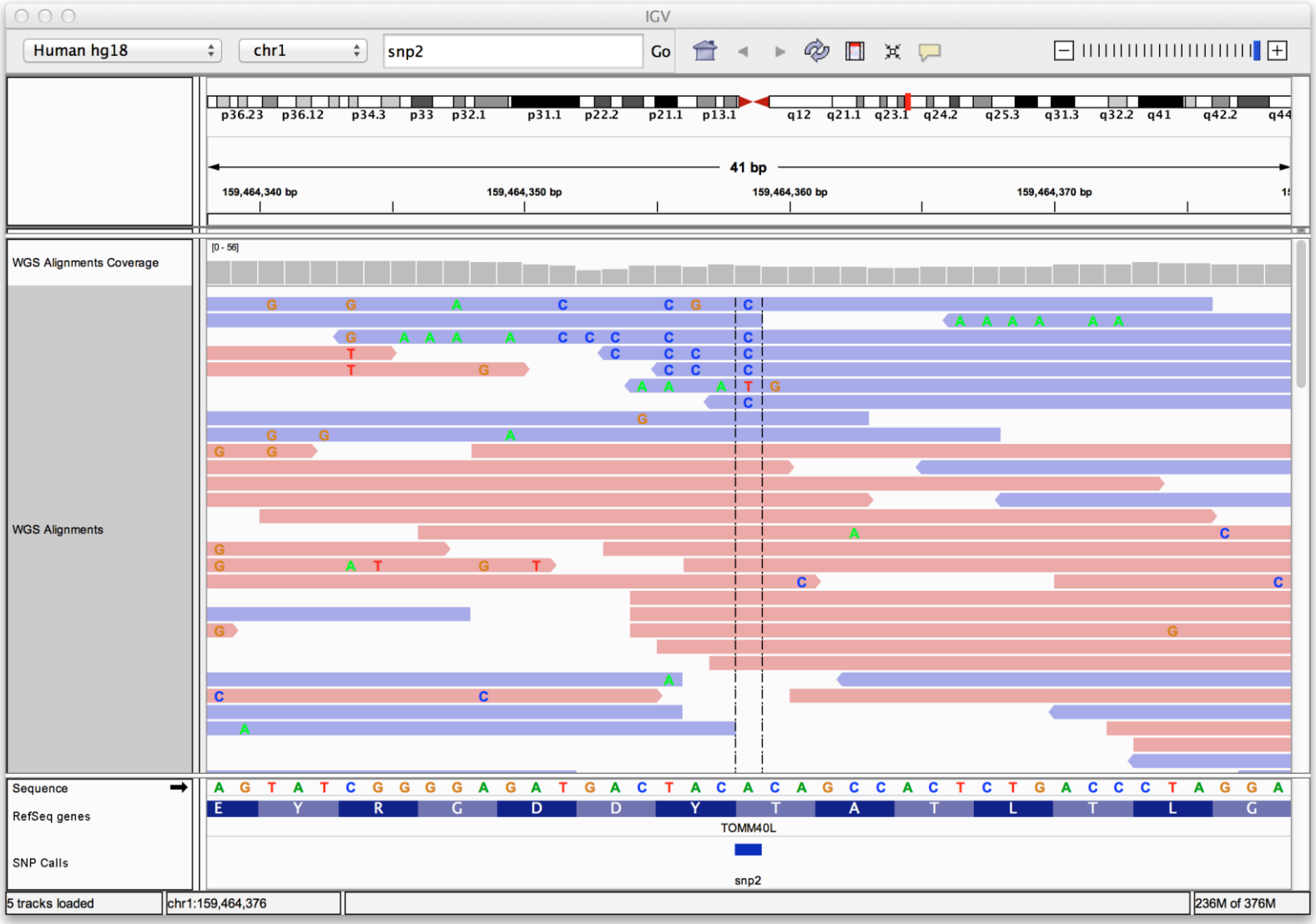
SNVs and Structural variations

- Important metrics for evaluating the validity of SNVs:
  - Coverage
  - Amount of support
  - Strand bias / PCR artifacts
  - Mapping qualities
  - Base qualities
- Important metrics for evaluating SVs:
  - Coverage
  - Insert size
  - Read pair orientation

Viewing SNPs and SNVs



Viewing SNPs and SNVs



Viewing Structural Events

- Paired reads can yield evidence for genomic “structural events”, such as deletions, translocations, and inversions.
- Alignment coloring options help highlight these events based on:
  - Inferred insert size (template length)
  - Pair orientation (relative strand of pair)

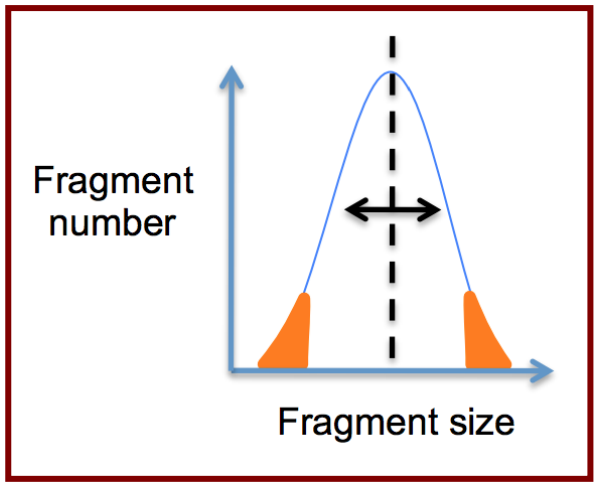
Paired-end sequencing

DNA or cDNA

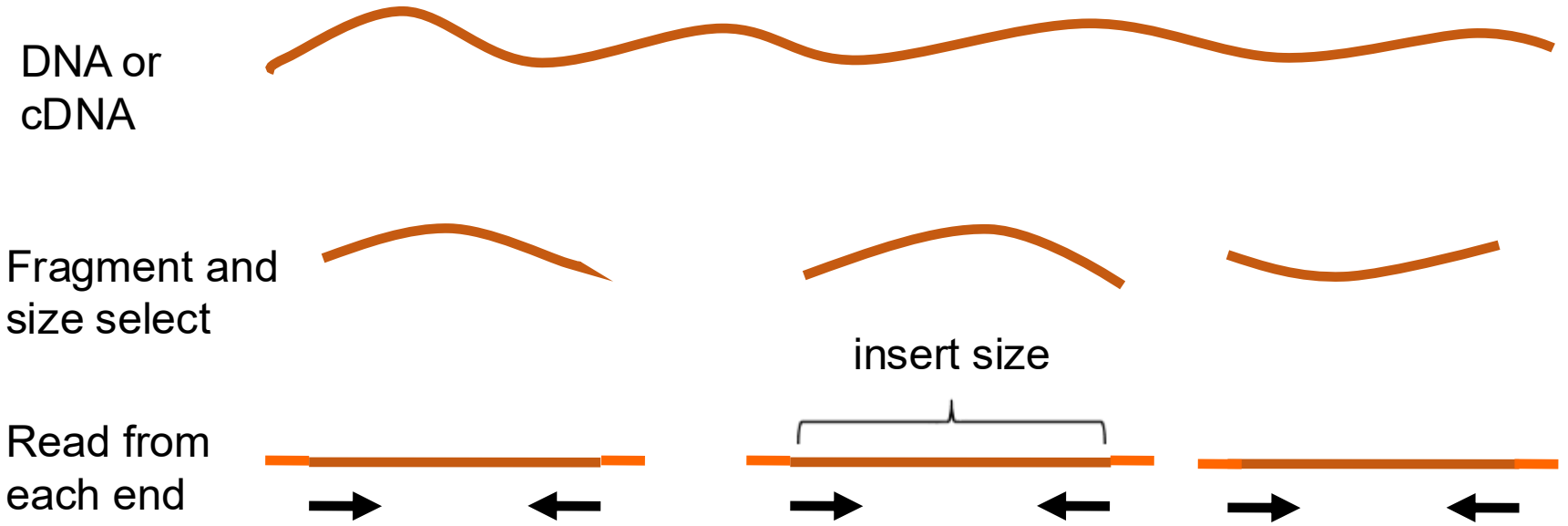


Fragment and size select

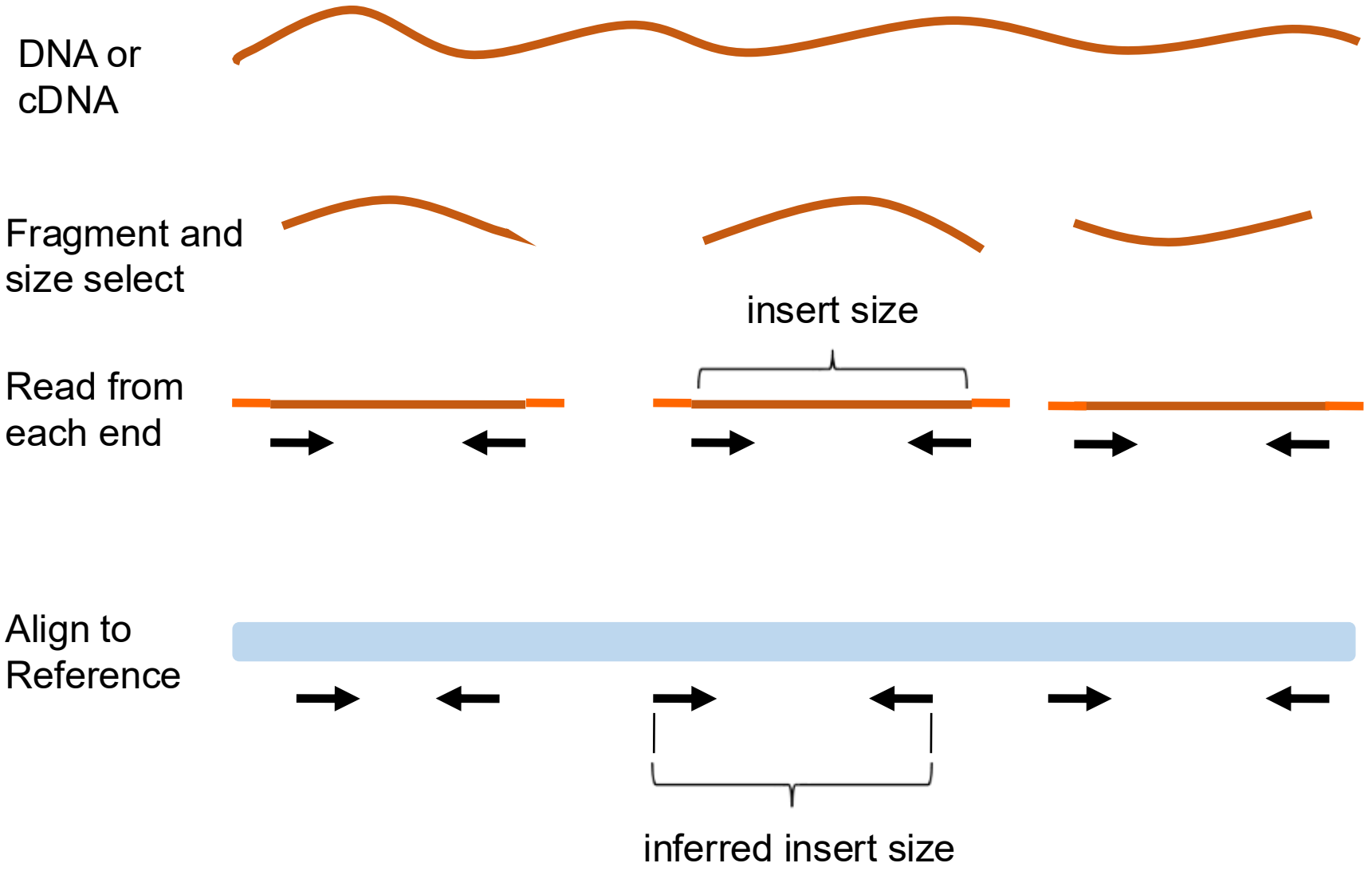




Paired-end sequencing



Paired-end sequencing



Interpreting inferred insert size

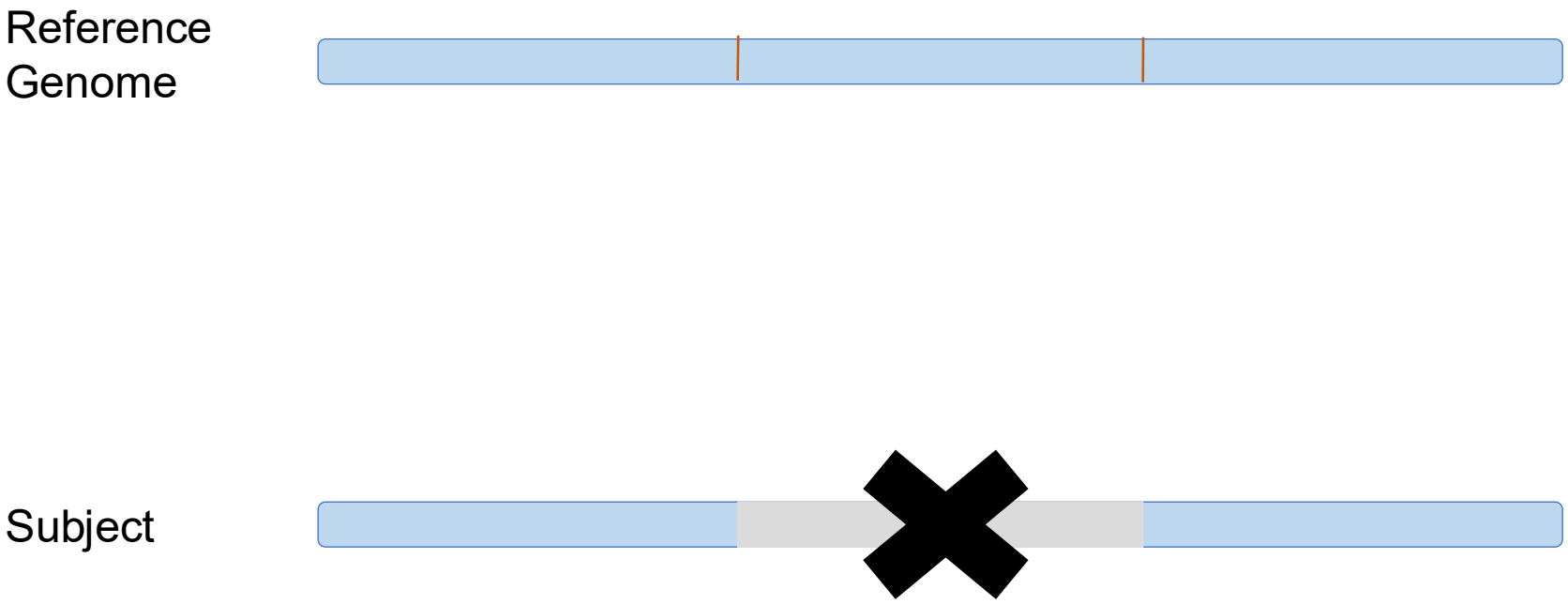
The “inferred insert size” can be used to detect structural variants including

- Deletions
- Insertions
- Inter-chromosomal rearrangements: (Undefined insert size)

Deletion

What is the effect of a deletion on inferred insert size?

Deletion



Deletion

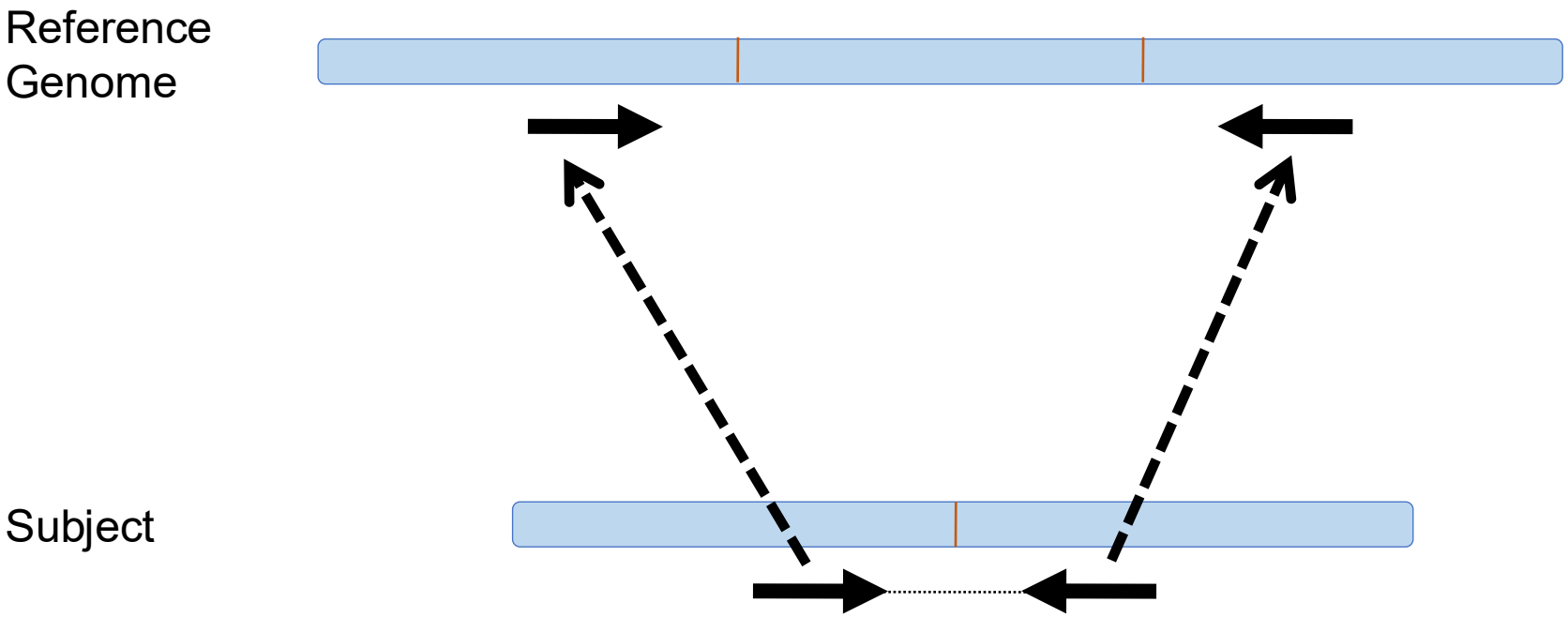
Reference Genome



Subject

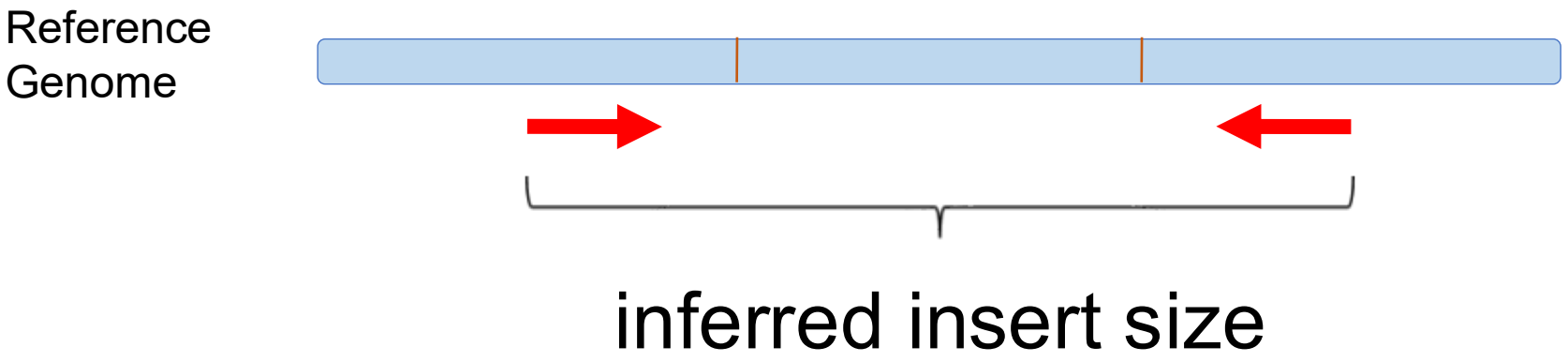


Deletion

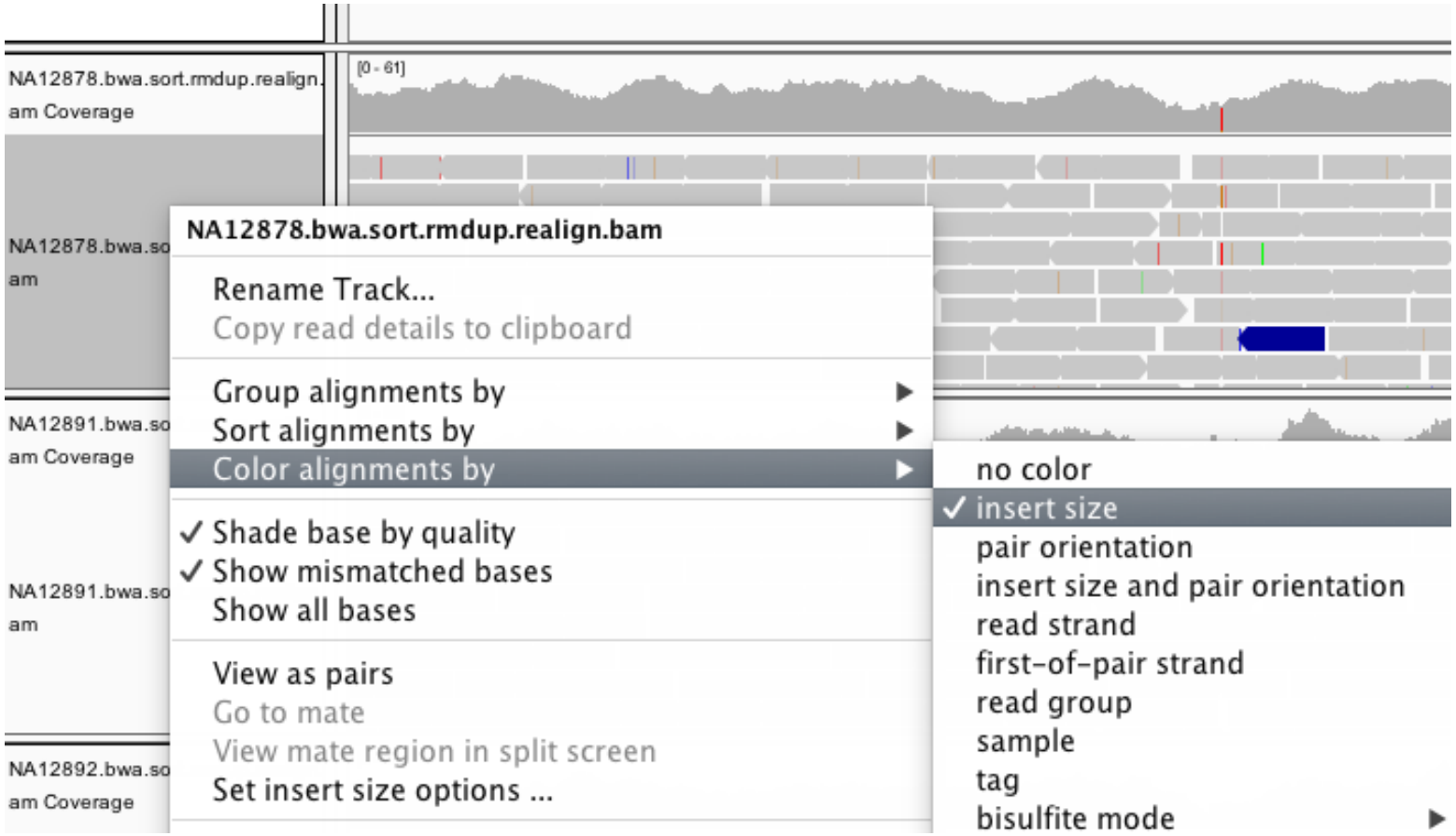


Deletion

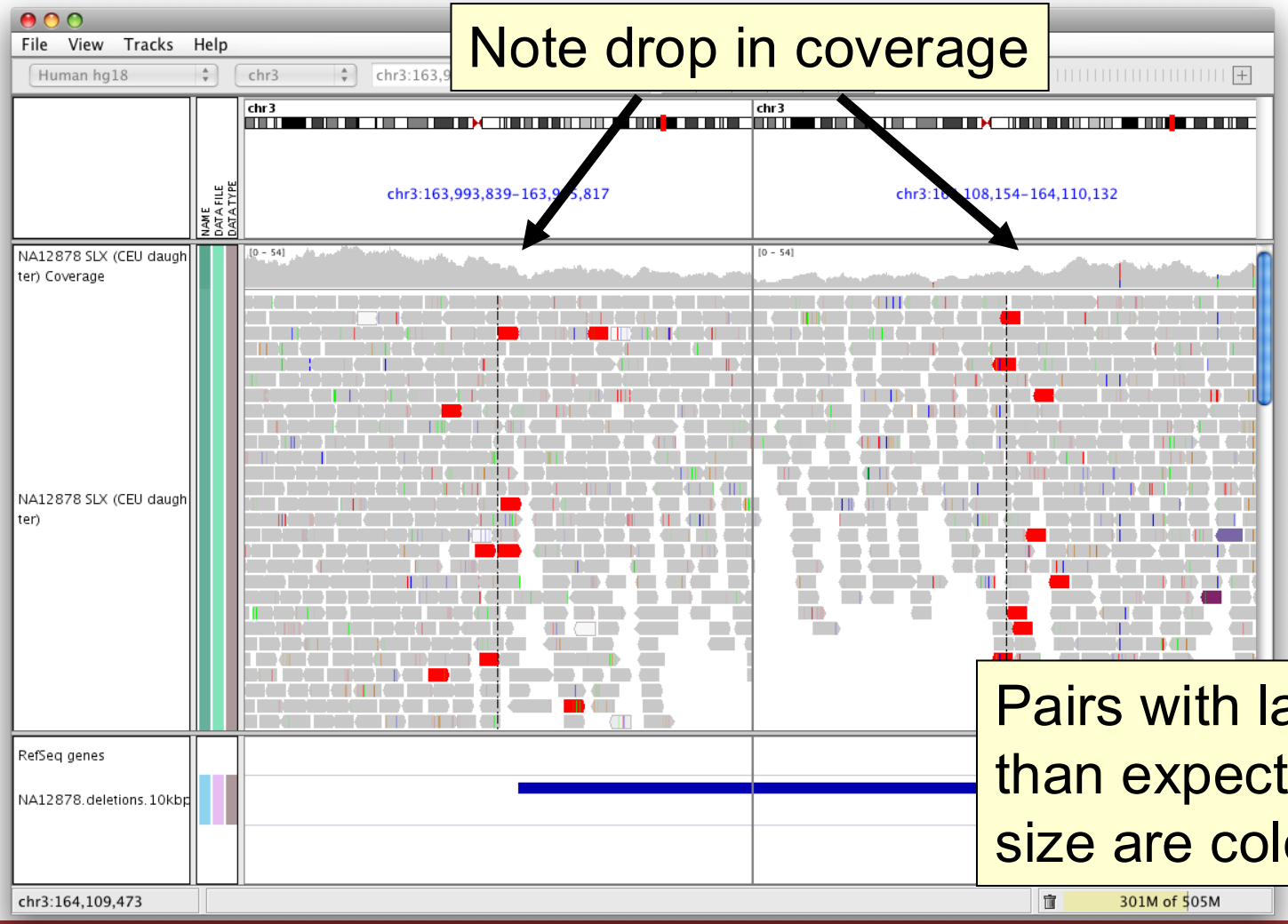
Inferred insert size is > expected value



Color by insert size



Deletion


Note drop in coverage
File View Tracks Help
Human hg18
chr3
chr3:163,9
chr3
chr3
chr3:163,993,839-163,9
chr3:1
108,154-164,110,132
NAME
DATA FILE
DATA TYPE
NA12878 SLX (CEU daugh ter) Coverage
[0 - 54]
[0 - 54]
NA12878 SLX (CEU daugh ter)
RefSeq genes
NA12878.deletions.10kbp
chr3:164,109,473
301M of 505M
Pairs with larger than expected insert size are colored red.

Insert size color scheme

- Smaller than expected insert size:



- Larger than expected insert size:

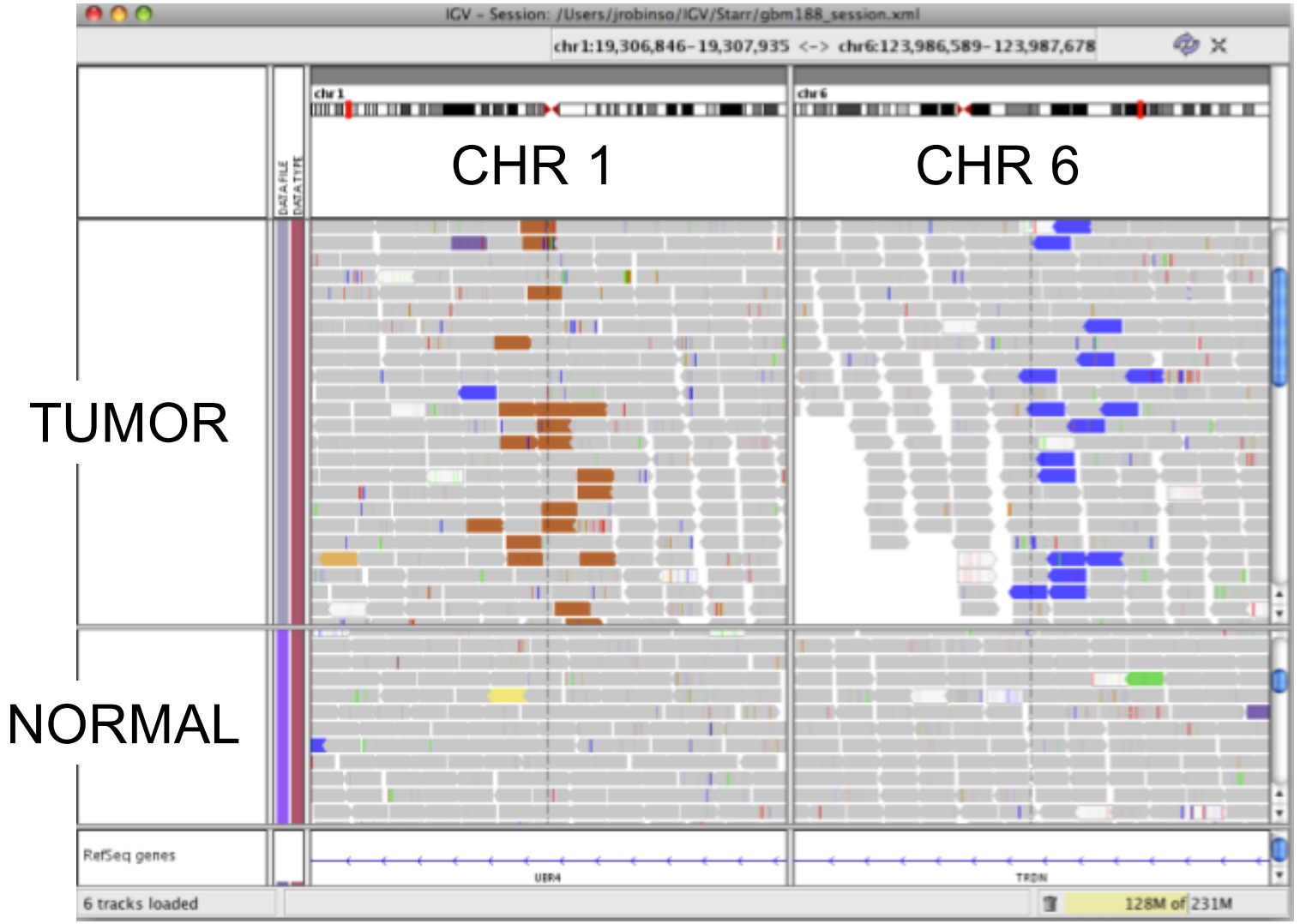


- Pairs on different chromosomes

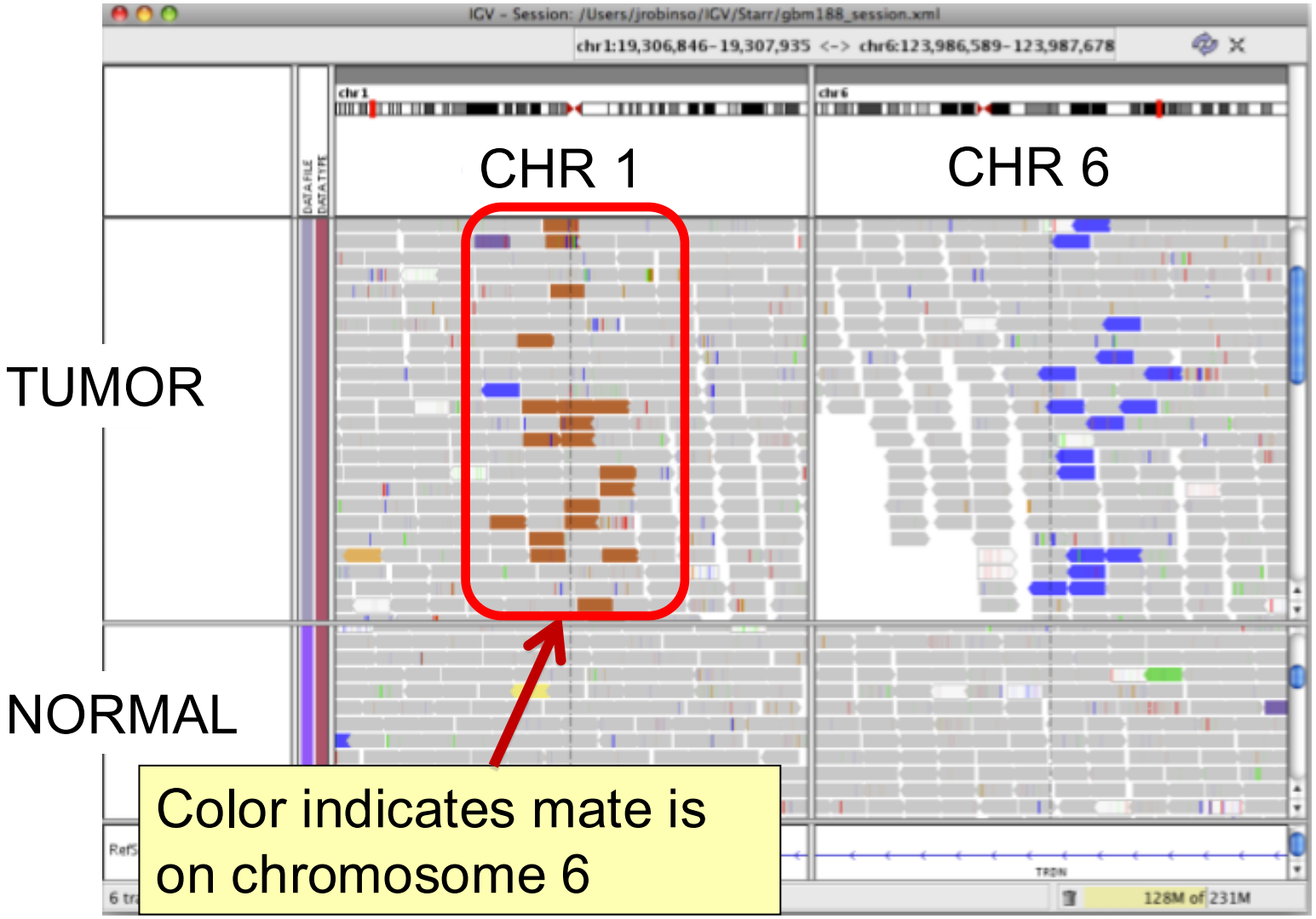
Each end colored by chromosome of its mate



Rearrangement



Rearrangement



Interpreting Read-Pair Orientations

Orientation of paired reads can reveal structural events:

- Inversions
- Duplications
- Translocations
- Complex rearrangements

Orientation is defined in terms of

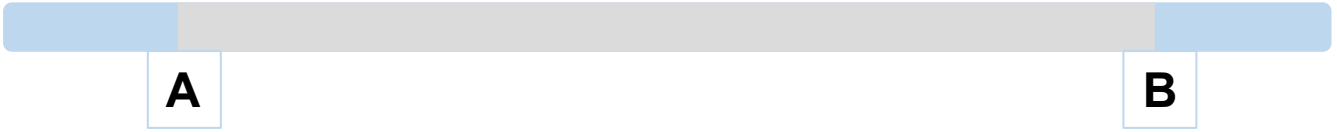
- read strand, left *vs* right, *and*
- read order, first *vs* second

Inversion

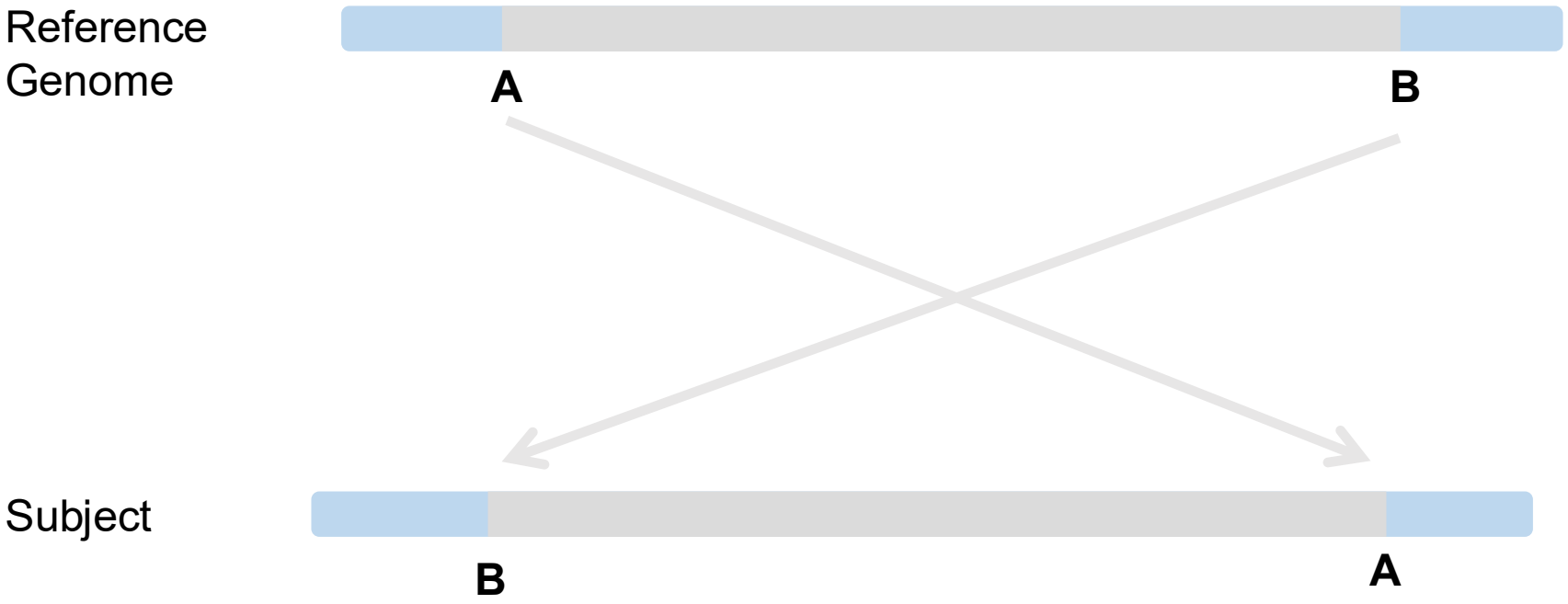
Reference genome

Inversion

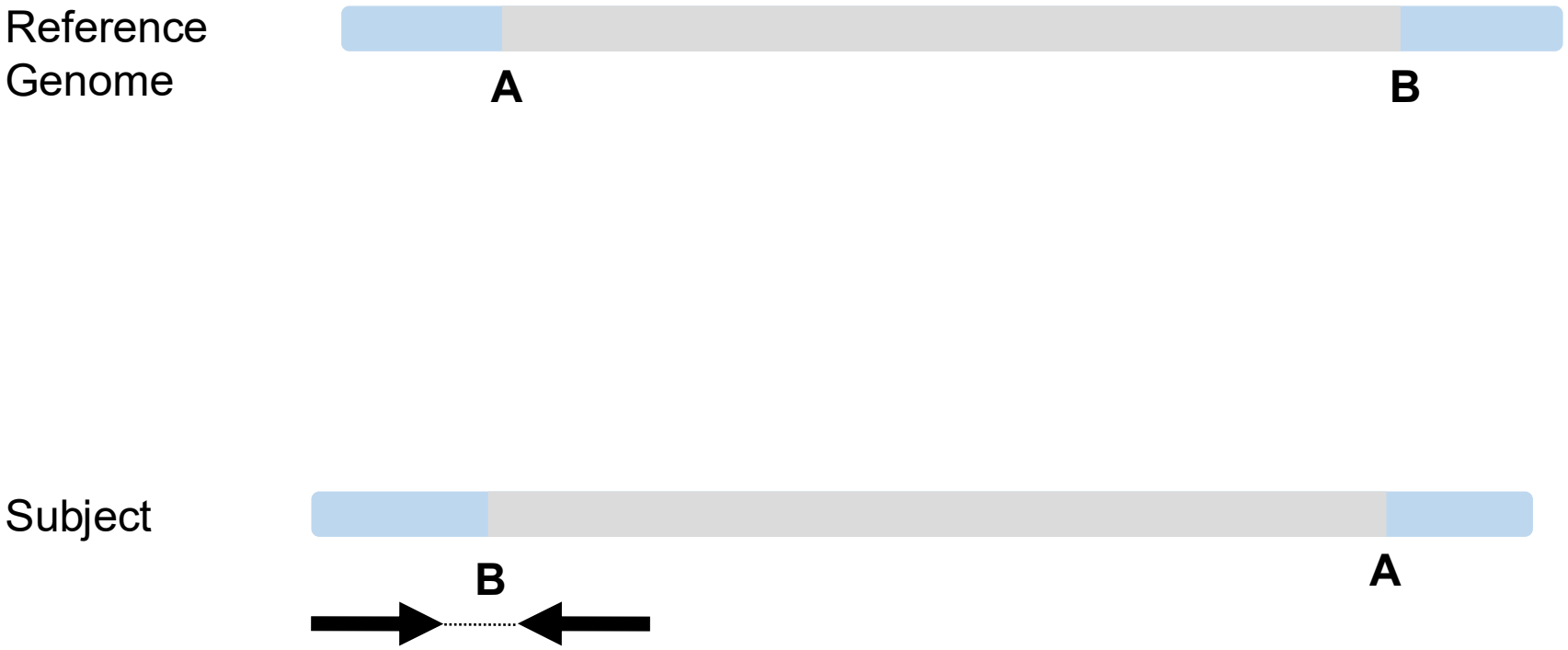
Reference genome



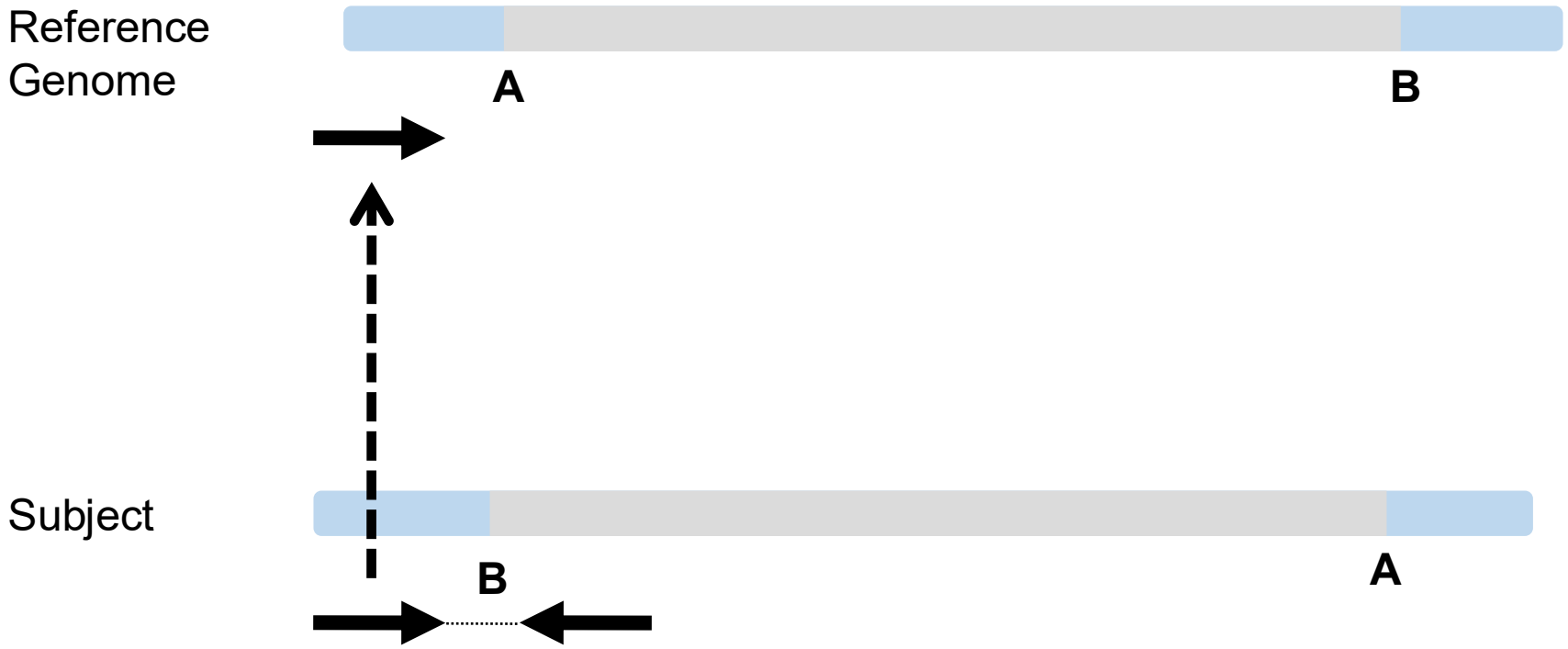
Inversion



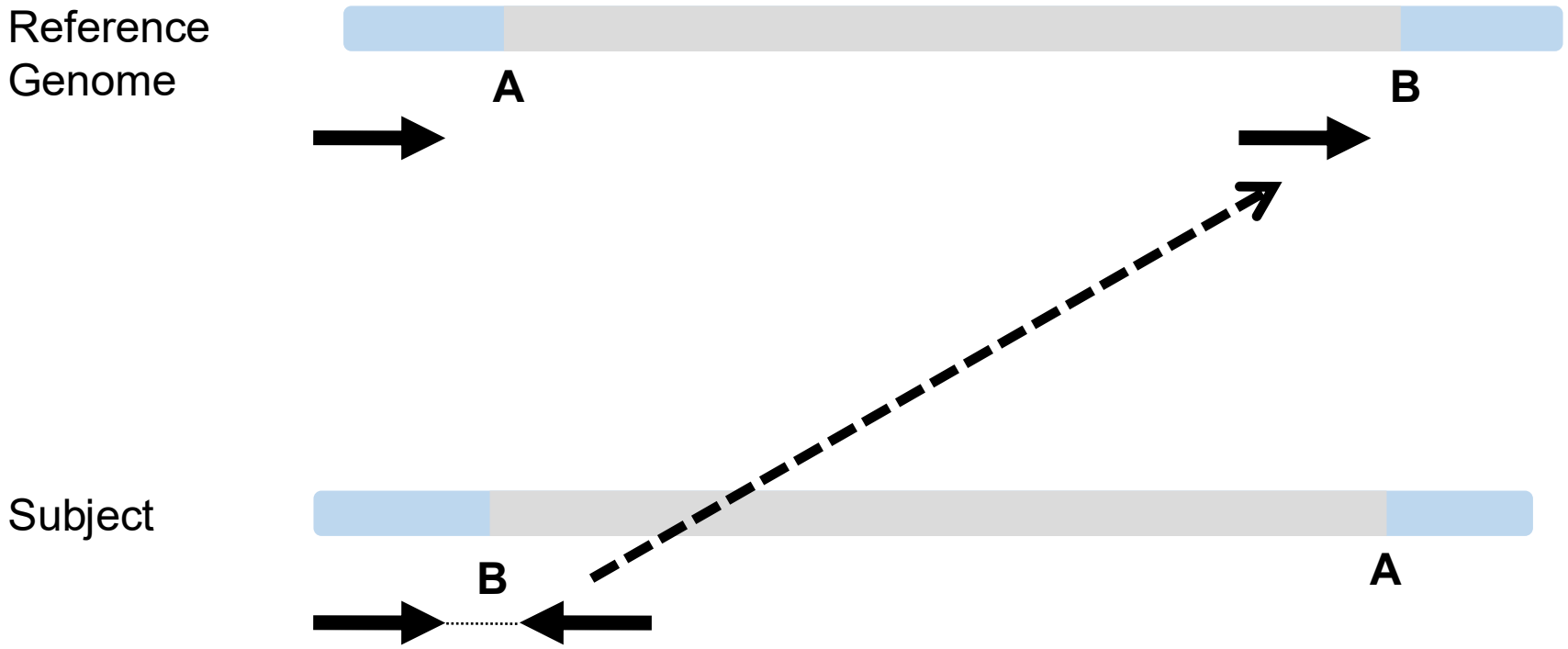
Inversion



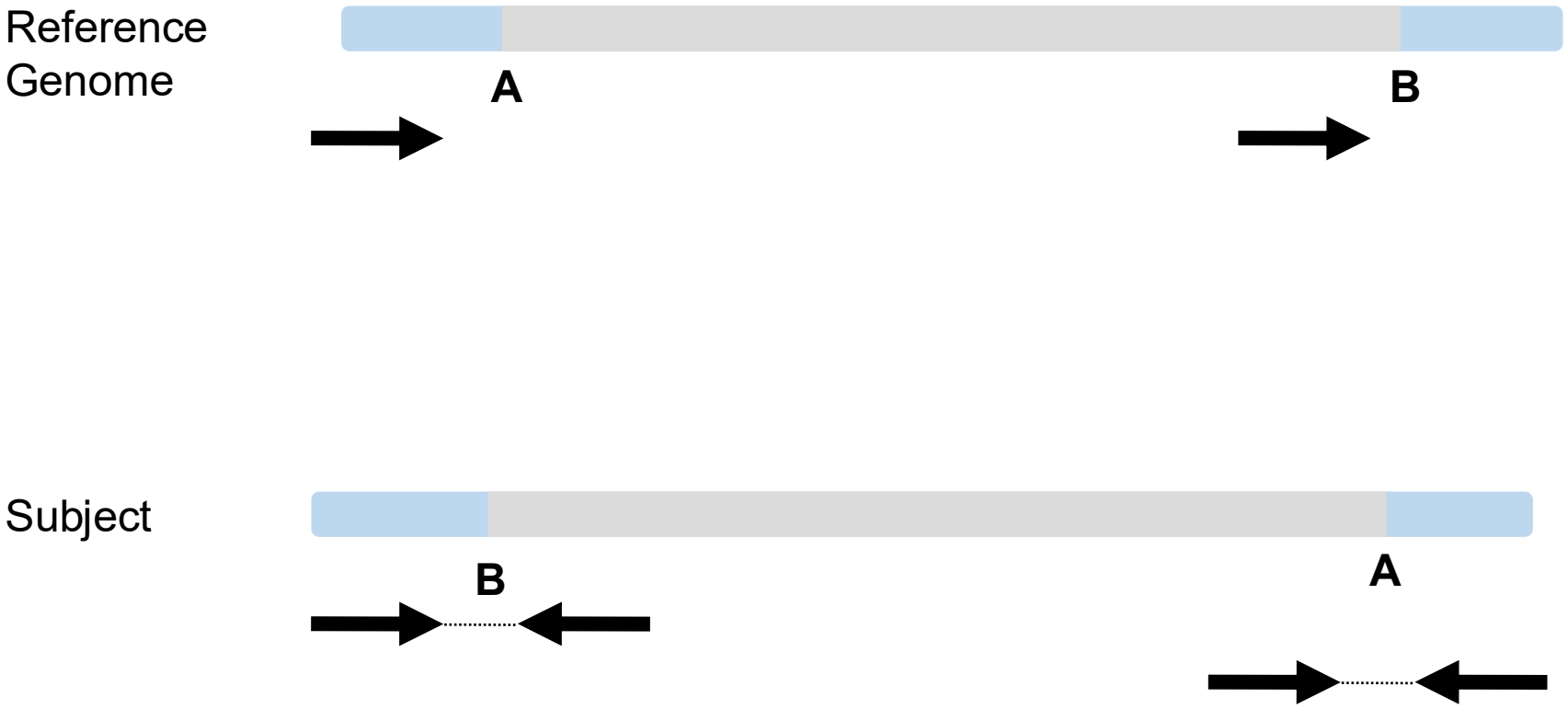
Inversion



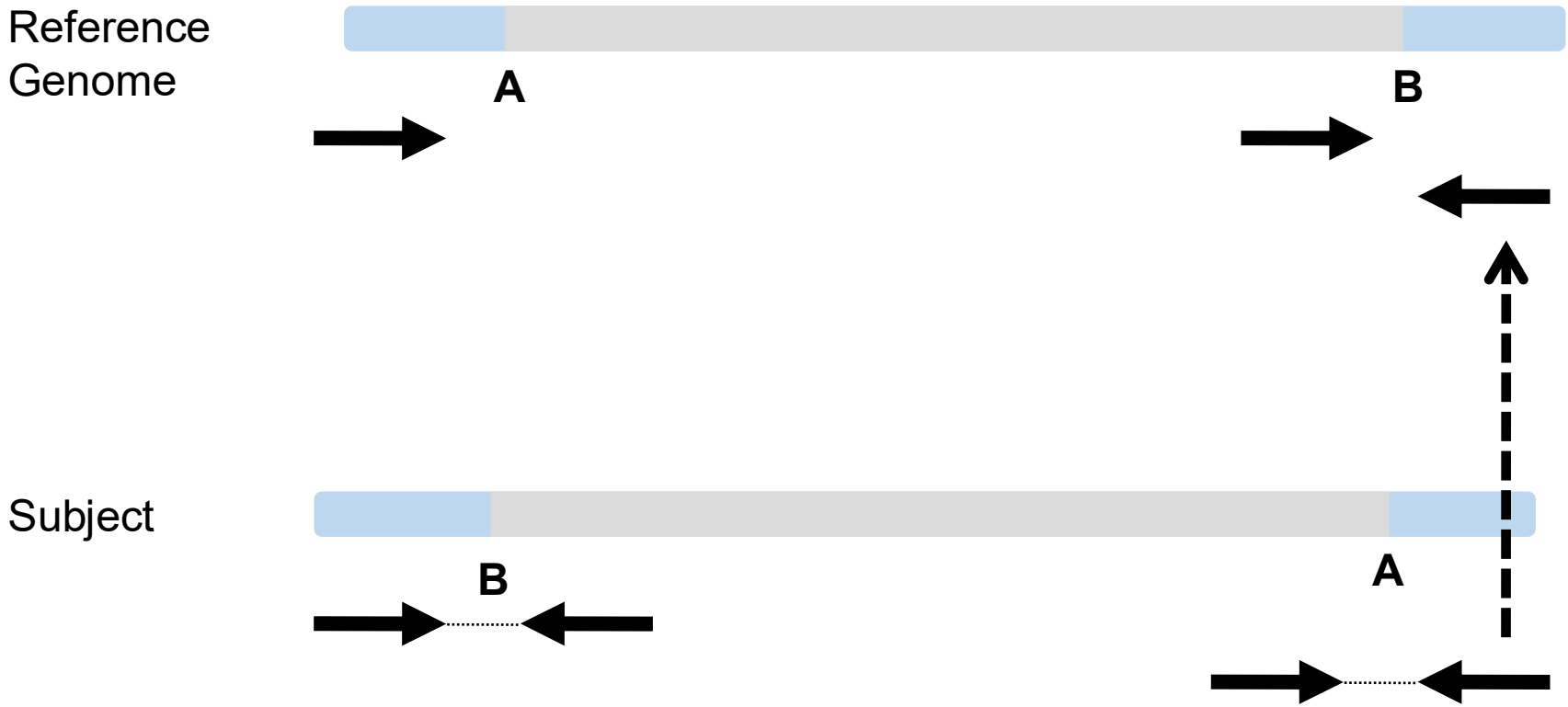
Inversion



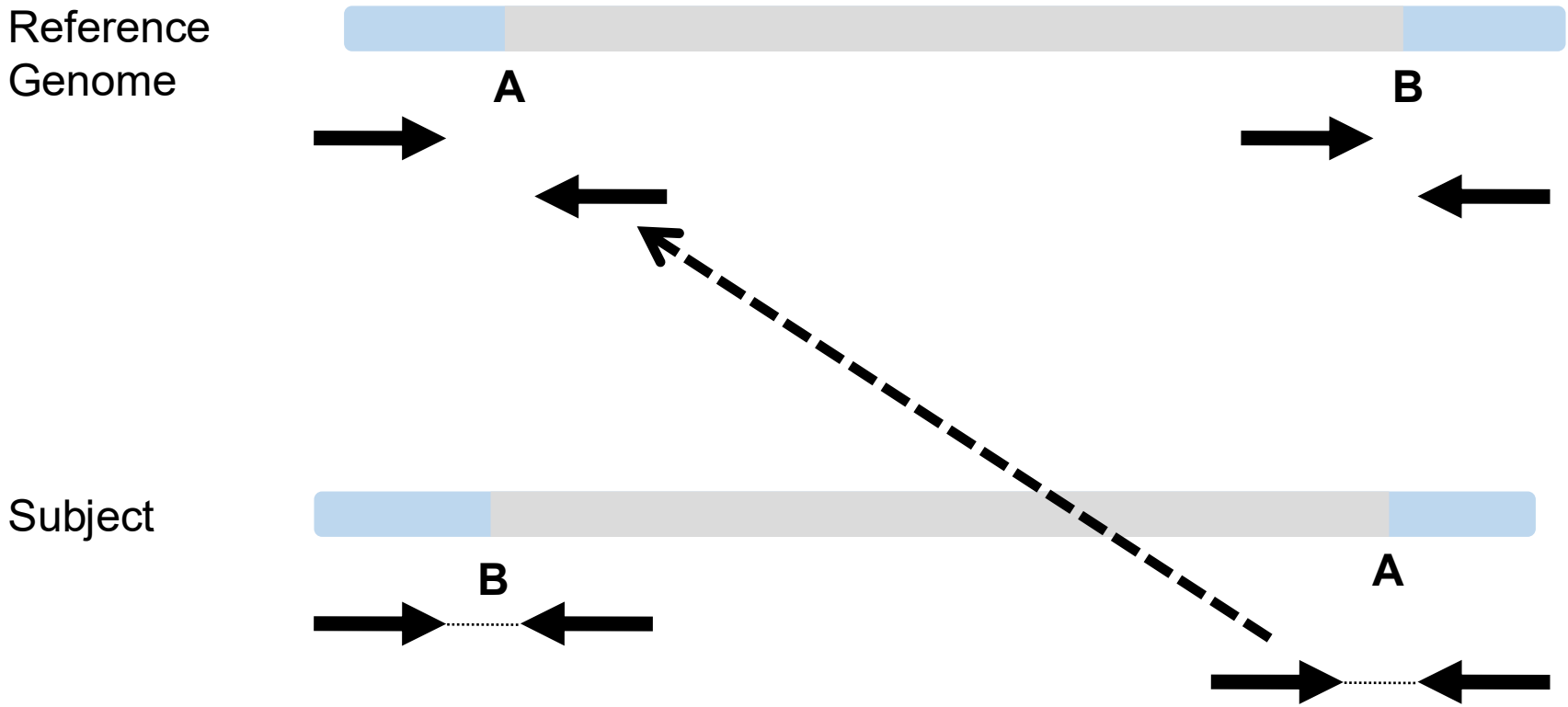
Inversion



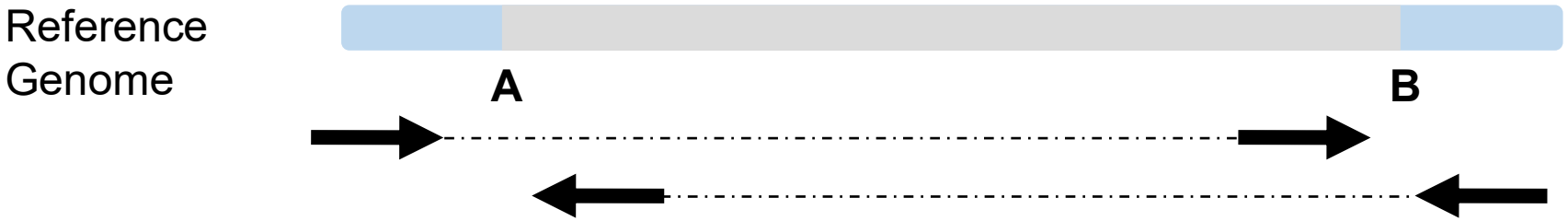
Inversion



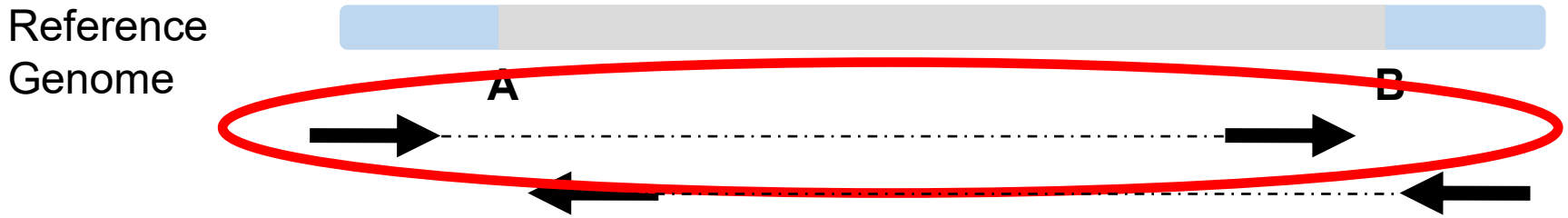
Inversion



Inversion

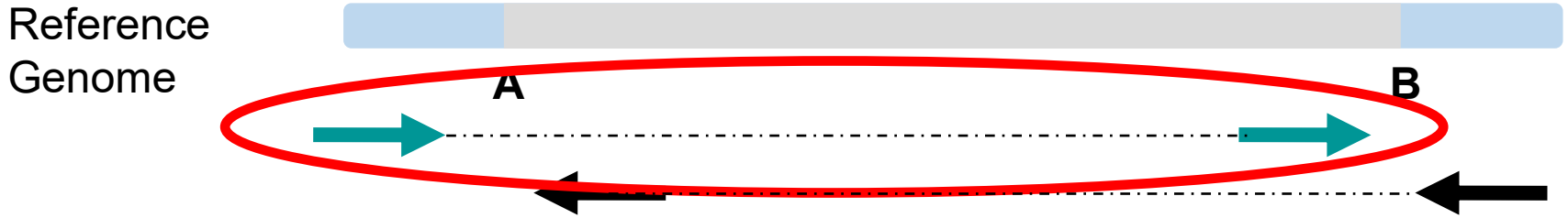


Inversion



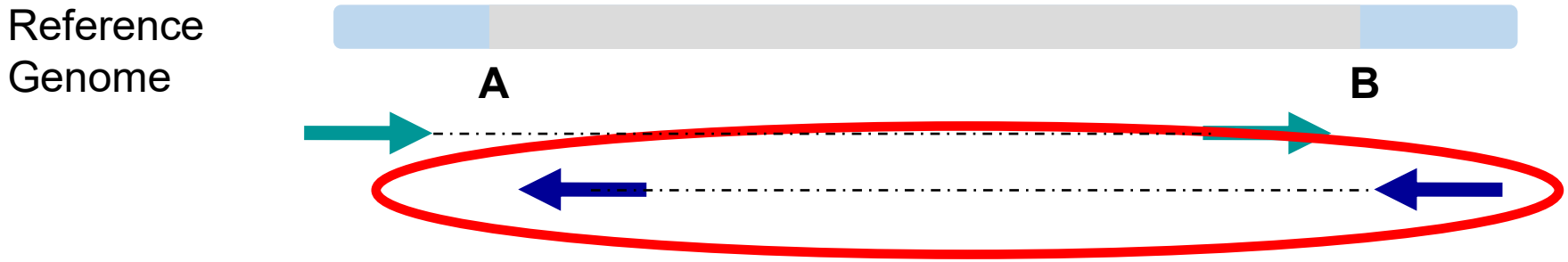
Anomaly: expected orientation of pair is inward facing (→ ←)

Inversion



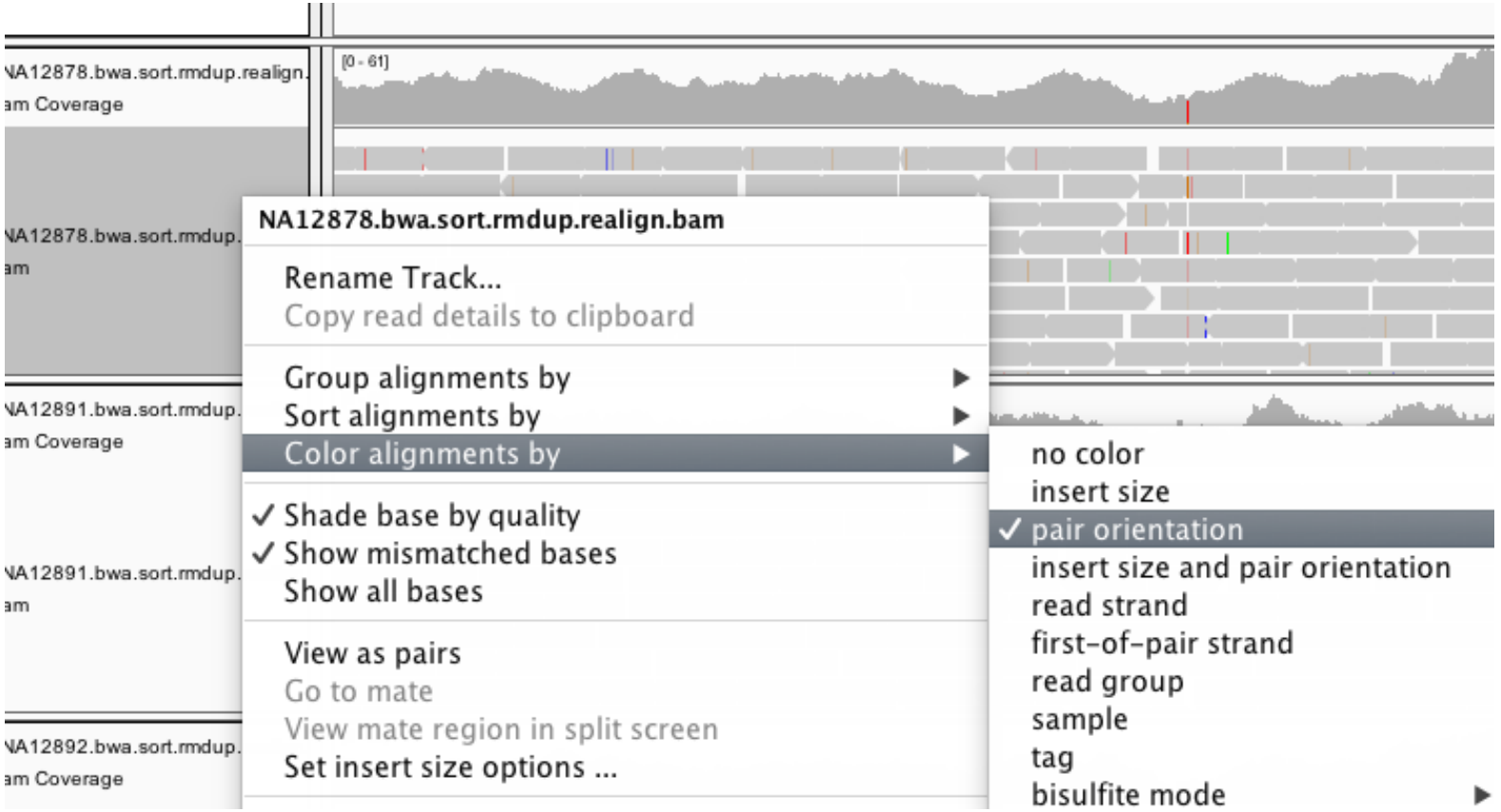
“Left” side pair

Inversion

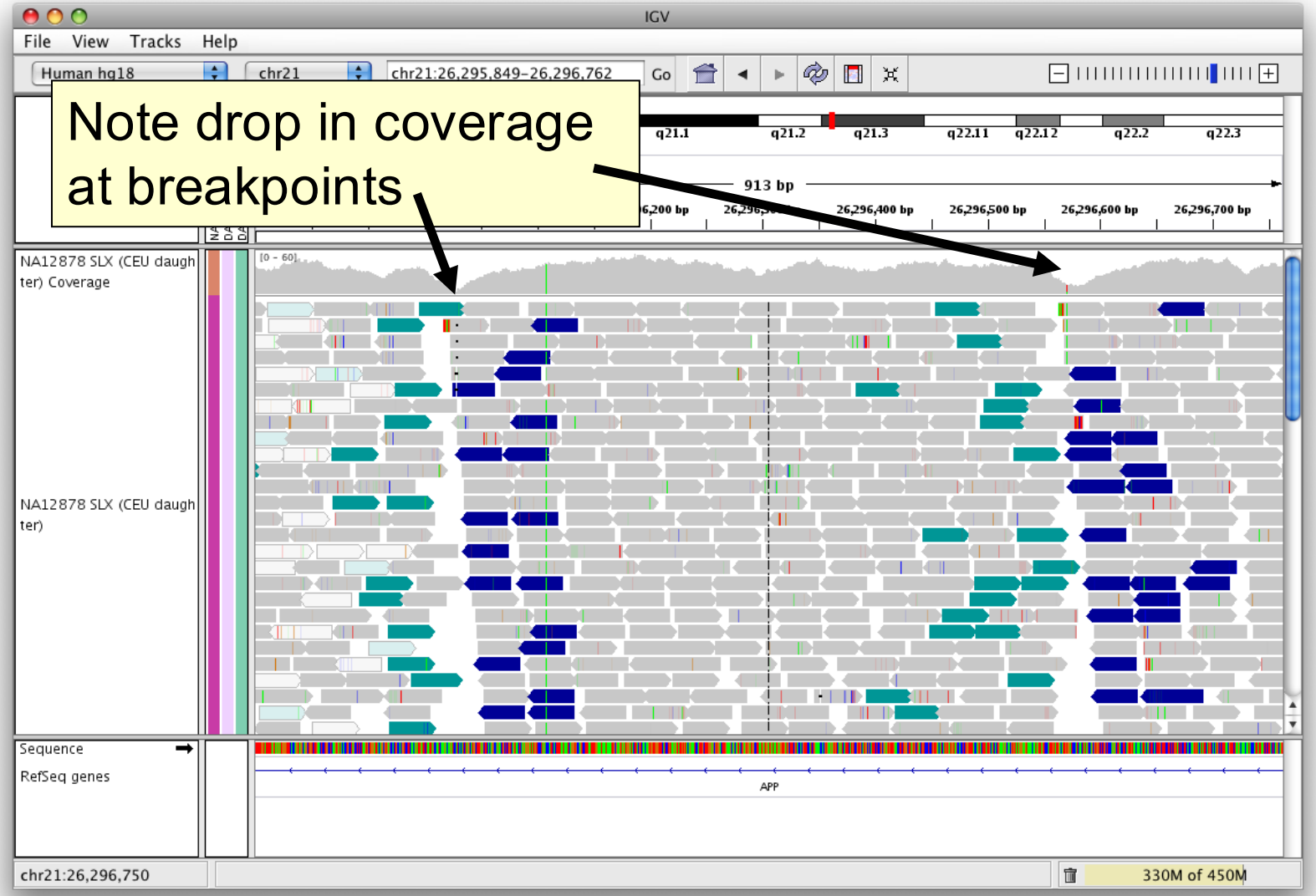


"Right" side pair

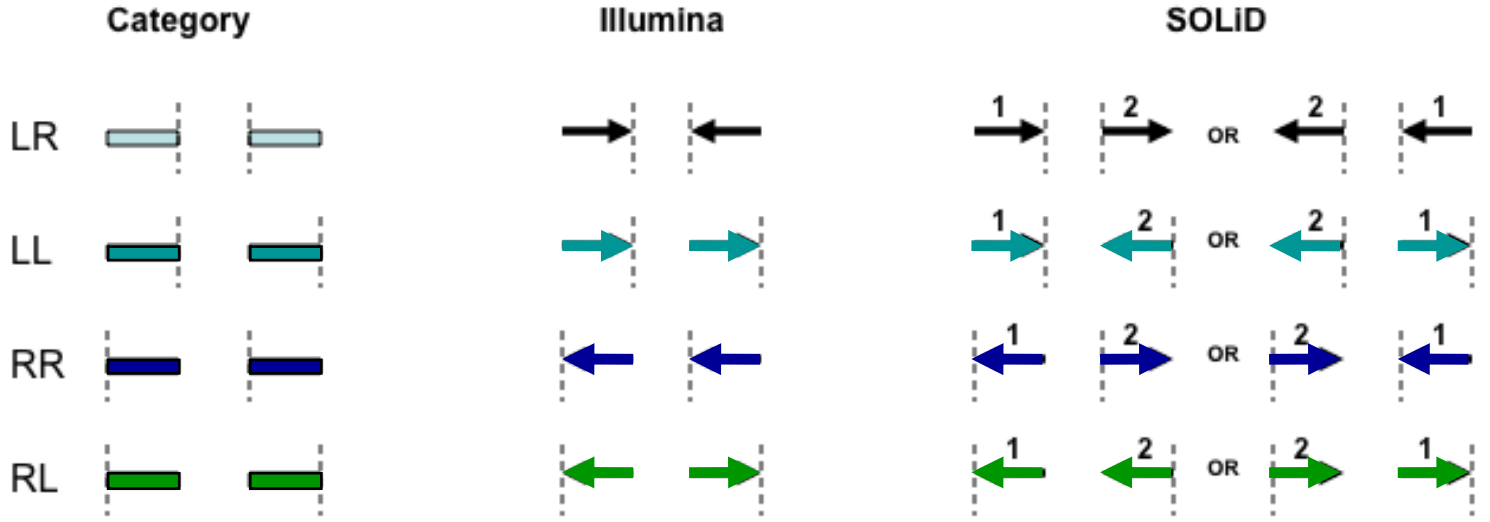
Color by pair orientation



Inversion


IGV
File View Tracks Help
Human hg18
chr21
chr21:26,295,849-26,296,762
Go
Note drop in coverage at breakpoints
913 bp
NA12878 SLX (CEU daughter) Coverage
NA12878 SLX (CEU daughter)
Sequence
RefSeq genes
APP
chr21:26,296,750
330M of 450M

Interpretation of read pair orientations



LR Normal reads.
The reads are left and right (respectively) of the unsequenced part of the sequenced DNA fragment when aligned back to the reference genome.

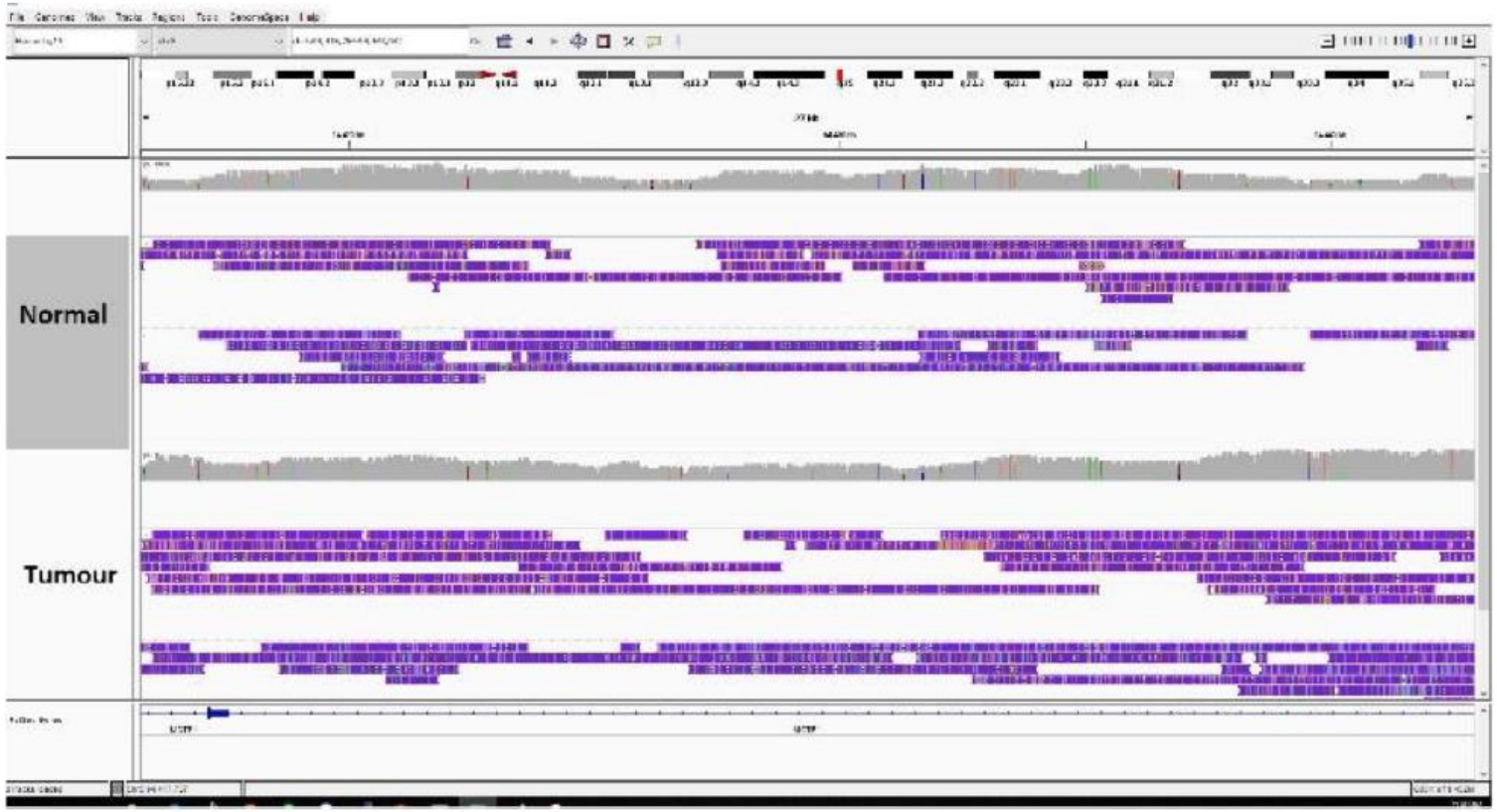
LL,RR Implies inversion in sequenced DNA with respect to reference.

RL Implies duplication or translocation with respect to reference.

These categories only apply to reads where both mates map to the same chromosome.

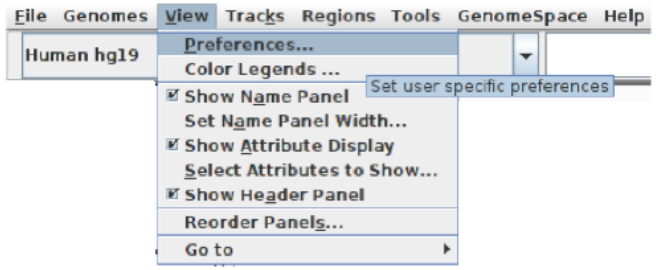
Figure courtesy of Bob Handsaker

Long read considerations

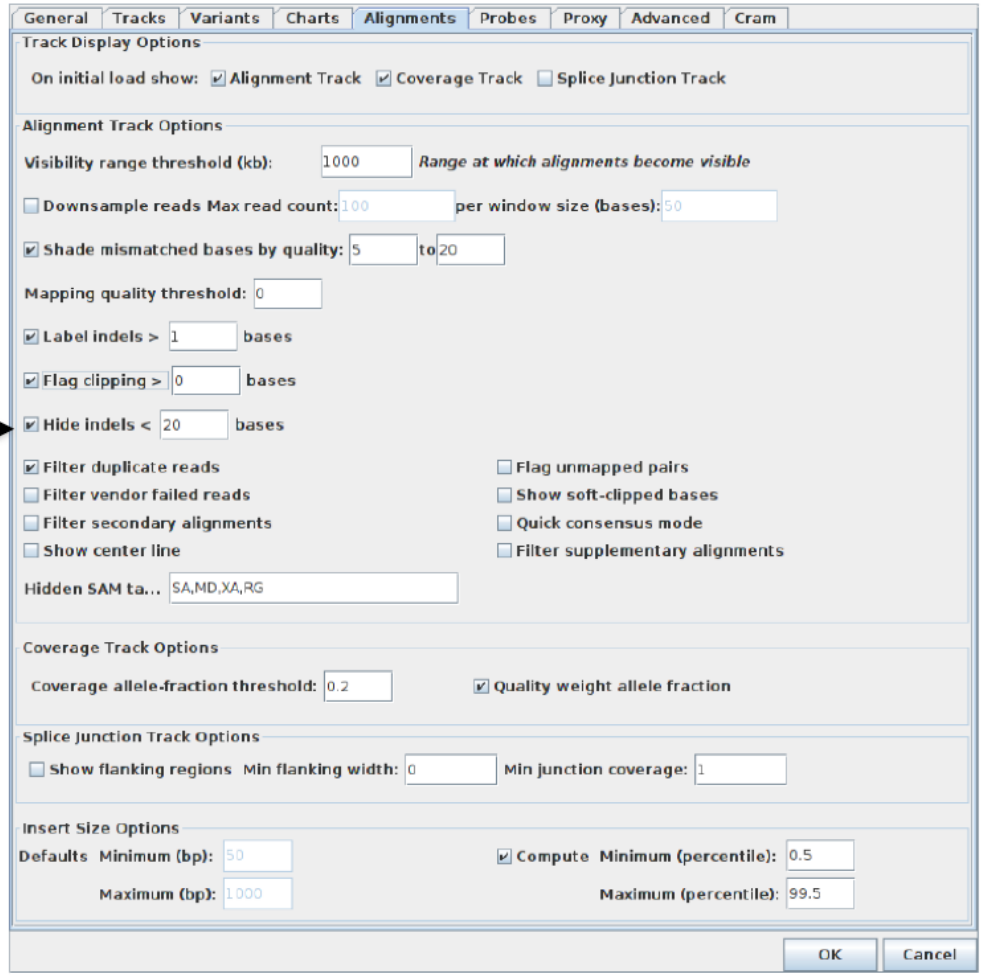


- Commonly see lots of small indels and single base errors that are simply noise
- Can be removed to be able to view the data more cleanly

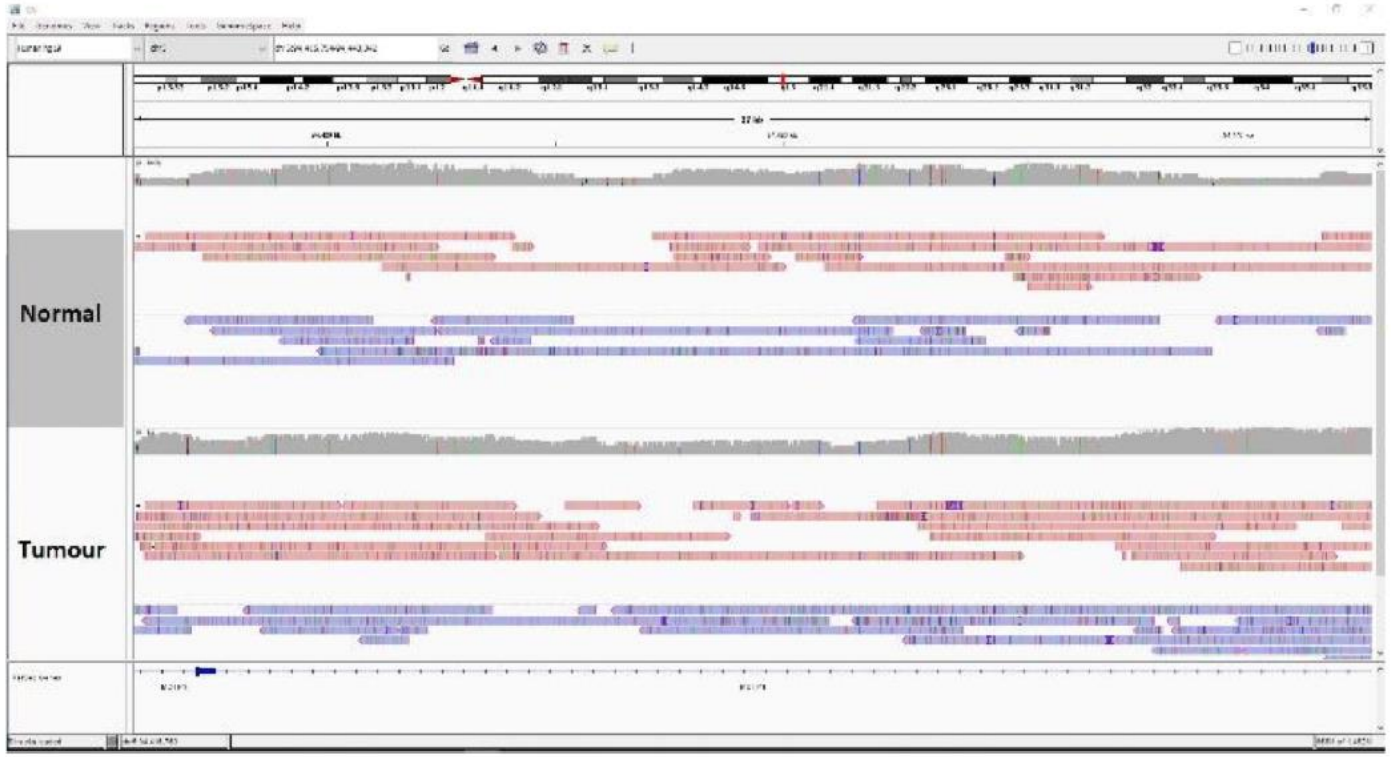
Long read considerations



Setting an indel threshold hides noise from small indels

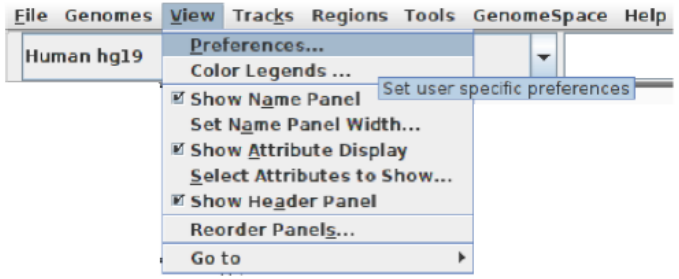


Long read considerations

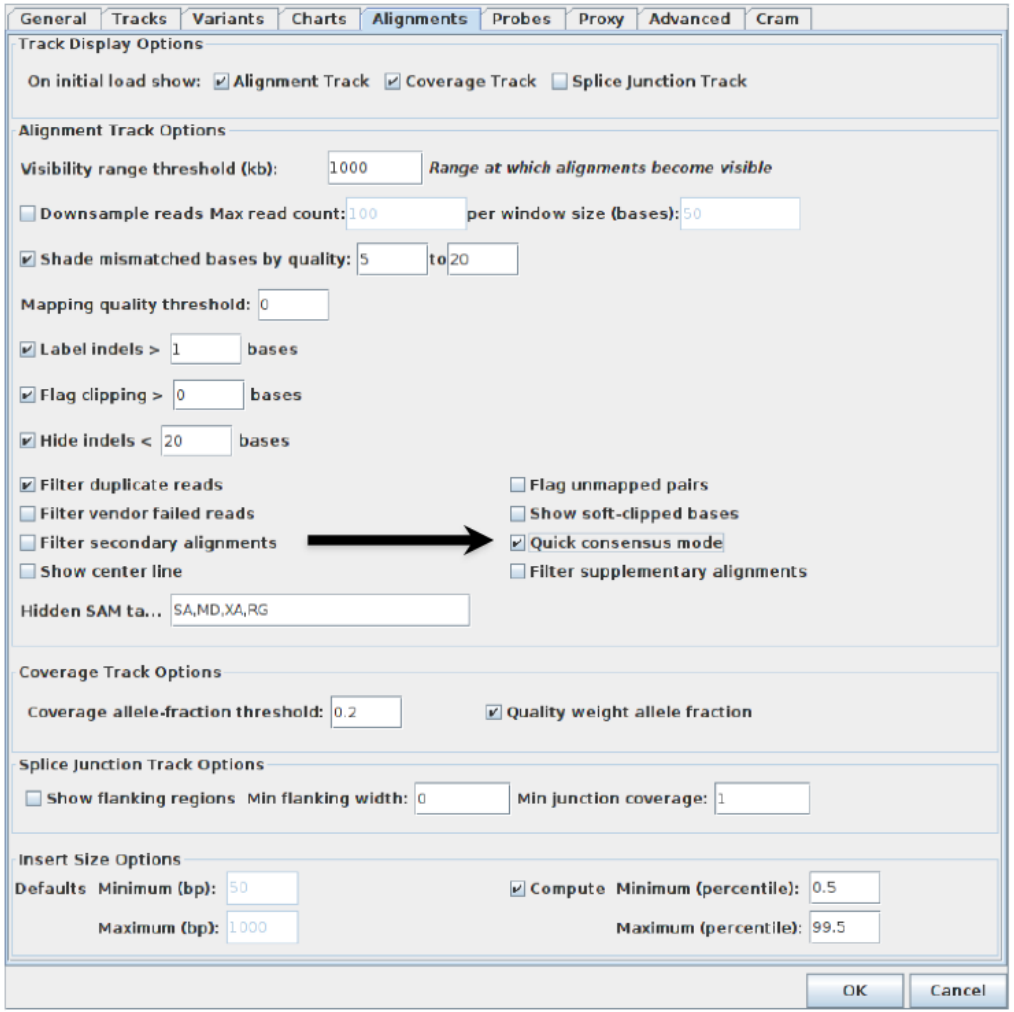


- Reads are not all purple dashes
- Next step would be to call a consensus at each position

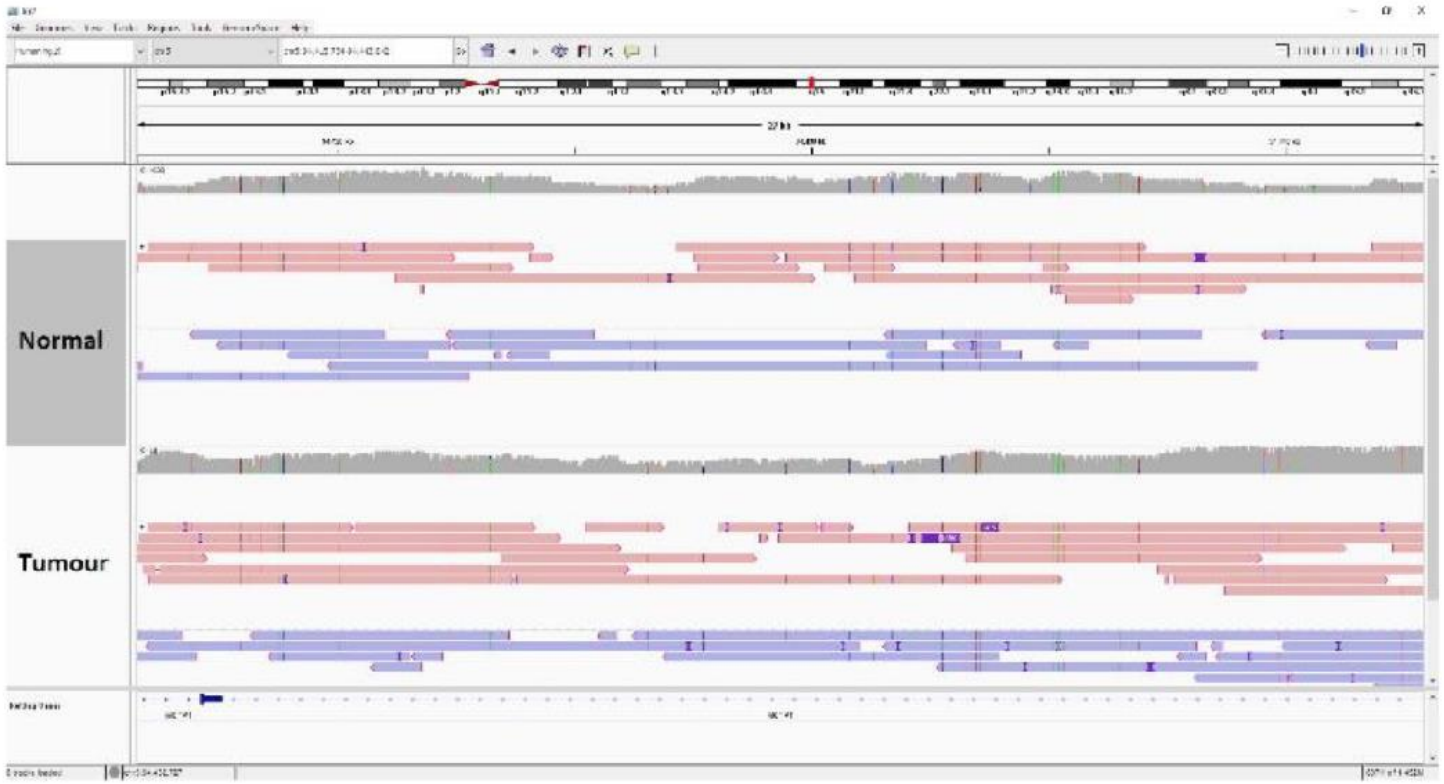
Long read considerations



Option for generating consensus sequences



Long read considerations



- Much easier to parse through the genomic data
- Large insertions and deletions are also labelled now

Manual Review Standard Operating Procedure (SOP) paper

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Standard operating procedure for somatic variant refinement of sequencing data with paired tumor and normal samples

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Purpose: Following automated variant calling, manual review of aligned read sequences is required to identify a high-quality list of somatic variants. Despite widespread use in analyzing sequence data, methods to standardize manual review have not been described, resulting in high inter- and intralab variability.

Methods: This manual review standard operating procedure (SOP) consists of methods to annotate variants with four different calls and 19 tags. The calls indicate a reviewer's confidence in each variant and the tags indicate commonly observed sequencing patterns and artifacts that inform the manual review call. Four individuals were asked to classify variants prior to, and after, reading the SOP and accuracy was assessed by comparing reviewer calls with orthogonal validation sequencing.

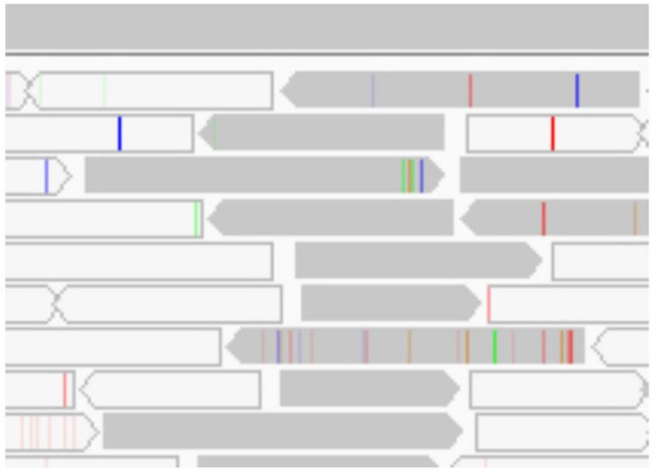
Results: After reading the SOP, average accuracy in somatic variant identification increased by 16.7% (p value = 0.0298) and average interreviewer agreement increased by 12.7% (p value < 0.001). Manual review conducted after reading the SOP did not significantly increase reviewer time.

Conclusion: This SOP supports and enhances manual somatic variant detection by improving reviewer accuracy while reducing the interreviewer variability for variant calling and annotation.

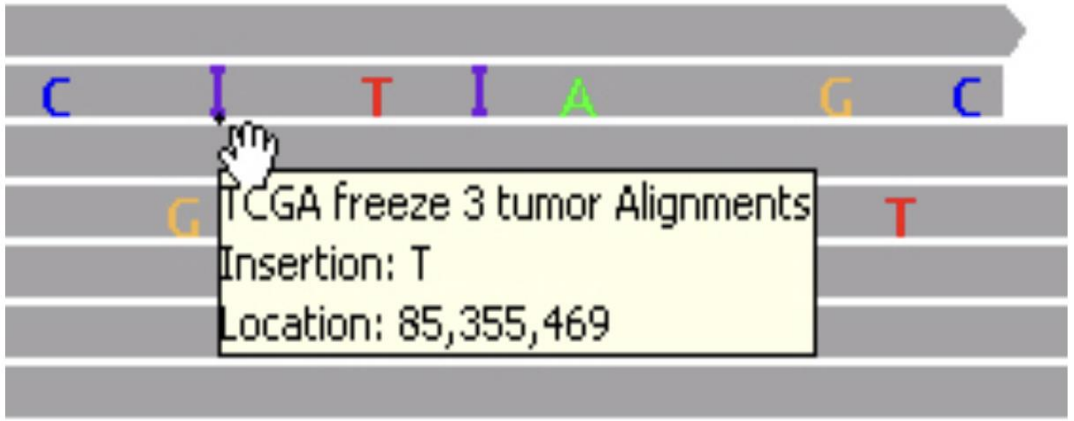
Genetics in Medicine (2018) https://doi.org/10.1038/s41436-018-0278-z

Keywords: somatic variant refinement; manual review

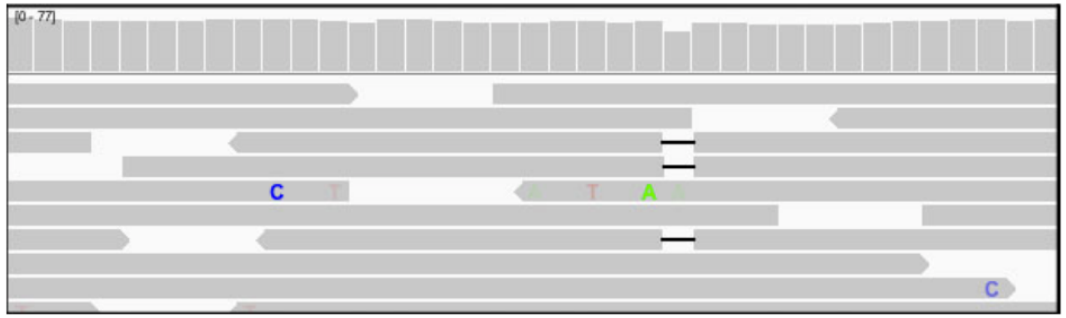
Other notes



Transparent (White) reads:
Low quality reads/
mapping quality equal to zero



Purple  : Insertion



Gapped read/black bar: Deletion

We are on a Coffee Break & Networking Session